

Introduction to Financial Markets – EFREI S7 2025

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Abstract

This course is conceived as a practical, broad-scope introduction to the key concepts and tools required for work in investment banking front office, quantitative analysis, or structuring. It is not intended to make students experts in finance, mathematics, or programming. Rather, it provides a hands-on overview of essentials in the following domains: financial markets and products, mathematical modeling, risk management, and Python programming, that practitioners in these fields must be at least familiar with. The aim is to equip students with a well-rounded “toolbox” that will help them navigate the practical challenges encountered in real-world front-office or quantitative environments. Depending on the specific applications encountered early in their careers, students will be able to deepen their knowledge of the topics introduced here to develop the specialized expertise required for their role. Note that over courses you might attend at EFREI will discuss theories covered in this course in greater detail: probability theory, mathematical modelling, Python core. The author has made every effort to combine theoretical insights with practical exercises. The hope is that students will derive maximum benefit from the material, not only by grasping the concepts presented, but also by feeling inspired to explore them further. Ultimately, the goal is for participants to leave with a solid foundation, a practical skill set, and the curiosity to continue expanding their expertise well beyond the classroom. On the following GitHub repository, you will find all the codes that are studied throughout this course: <https://github.com/SK8gh/EFREI/tree/main>



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1 Introduction to Financial Markets and Products

Financial markets are central to the functioning of the global economy. They act as meeting points for investors and institutions, where financial instruments are exchanged.

These markets can be classified based on their function - primary markets for issuing new assets, and secondary markets for trading existing ones - or by the type of asset involved.

Efficient markets are those that are transparent, well-regulated, and structured to protect participants. This environment helps channel savings into productive investments, spread financial risks, and support innovation and economic expansion.

Institutions such as banks, asset managers, insurance firms, and investment funds play a crucial role in moving capital through the system. They help transform savings into credit and investments, and offer financial services tailored to both individuals and corporations.

Assets themselves represent financial claims - equities reflect company ownership, bonds are loan agreements with regular repayments, currencies enable international trade, and derivatives allow investors to manage uncertainty or take positions based on expected price movements.

Oversight of the financial system is handled by regulatory entities like central banks and financial authorities. Their role is to maintain market order, enforce legal standards, safeguard consumers and clients, and ensure confidence in the financial system.

Careers in finance are diverse, offering a range of opportunities depending on skills, interests, and preferred working style. Some of the main roles include:

- **Trader:** A front-office professional responsible for taking positions in financial markets by buying and selling instruments such as equities, fixed income, derivatives, foreign exchange, etc... Traders manage the *risk-reward profile* of their books, monitor market movements in real time, execute client or proprietary orders, and optimize profitability while respecting risk limits. They use pricing models, market intuition, and flow information to make decisions. Traders are usually specialized by asset class (e.g. FX, credit, commodities) and strategy (market-making, proprietary trading, arbitrage).
- **Sales:** The client-facing arm of the front office, sales professionals maintain relationships with institutional clients (asset managers, hedge funds, corporations, pension funds). They understand client needs, present investment ideas, market updates, and structured solutions, and act as intermediaries between clients and traders. They must balance *commercial objectives* (building revenue and wallet share) with providing *best execution and relevant advice*, often requiring deep product knowledge and communication skills.
- **Structurer:** Specialists who design and customize complex financial products, such as structured notes, exotic derivatives, or tailored hedging

strategies. They combine market insight, quantitative modeling, and knowledge of legal/tax frameworks to create solutions aligned with client objectives (yield enhancement, risk hedging, exposure creation). Structurers sit at the intersection of *trading, sales, and quantitative research*, translating market views and client constraints into executable deal structures. They often prepare *pitchbooks* and *term sheets* with precise payoff diagrams.

- **Quantitative analyst (Quant):** Experts in mathematical modeling and numerical methods applied to finance. Quants develop pricing models for derivatives, risk measures (VaR, Greeks, XVA), and algorithmic strategies. They implement these models in programming languages (Python, C++, Java, or increasingly Rust) and collaborate with traders and structurers to ensure models are both mathematically robust and computationally efficient. Their work spans from *research and prototyping* to *production implementation* in trading systems, requiring strong backgrounds in probability theory, stochastic calculus, and numerical optimization. Note that, even in quant positions, it is very rare to actually do math, create models, numerical methods, etc... It is more and more about programming in a very functional environment, driven by business incentives.
- **Risk analyst:** Professionals ensuring that the risks taken by trading desks and portfolios are *identified, measured, and controlled*. They monitor daily exposure to market risk (delta, gamma, vega, VaR), credit risk (counterparty exposure, default probabilities), and liquidity risk. Risk analysts produce daily reports, stress-test portfolios under extreme scenarios, and verify compliance with regulatory frameworks (Basel, MiFID, MAD-MAR, etc...). They also challenge traders' assumptions and escalate breaches of risk limits to senior management. Their role is to balance business profitability with financial stability and regulatory safety.
- **Middle office professional:** bridges the front and back offices, including trade support, portfolio control, and risk reporting, ensuring accurate transaction capture, correct risk measurement, and regulatory compliance.
- **Back office and operations professional:** provides essential infrastructure, including clearing, settlement, accounting, and IT systems, enabling the firm to function smoothly and securely.

1.1 History of major crashes: 1637, 1929, 1987, 2008

As an introduction to financial markets, it is instructive to revisit some of the major crises that have marked their history. Throughout the past four centuries, episodes of speculative excess and subsequent collapse have not only disrupted economies, but also reshaped the very structure of markets, regulation, and financial theory. Studying these events allows us to understand how bubbles form, how systemic risks propagate, and why each crash has left behind lessons that continue to influence modern finance.

"Those who fail to learn from history are condemned to repeat it"

Tulpenmanie - 1637

Tulpenmanie² occurred in the Dutch Republic. Initially a luxury good for the wealthy, tulip bulbs – especially rare varieties like the *Semper Augustus* – turned into a speculative asset, they became a status symbol.

Prices soared: some bulbs reportedly sold for over 5.000 guilders³, equivalent to the cost of a canal house in Amsterdam or more than ten times a skilled worker's annual income.

Speculation was fueled by informal forward contracts, often referred to as *windhandel*⁴.

These were agreements to buy bulbs at a future date and price, typically settled in the spring when bulbs would be lifted from the ground. Crucially, these contracts were not standardized, not traded on any formal exchange, and did not involve clearinghouses⁵ or margin calls⁶.

They were rarely backed by collateral, and only a small deposit - about 2.5% of the agreed price was required to secure the deal⁷. Moreover, these agreements were not legally binding in civil courts, which meant buyers could walk away if prices moved unfavorably.

Unlike modern futures contracts, which are standardized, regulated, and legally enforceable through centralized exchanges and clearinghouses, the tulip contracts lacked enforceability, transparency, and risk management mechanisms. There was no obligation for physical delivery, meaning no actual bulbs needed to change hands unless both parties chose to settle.

This allowed contracts to be traded purely for speculative gain, sometimes changing owners multiple times before the delivery date, with no real connection to a need in the underlying asset.

The bubble burst in early February 1637 when a routine auction in Harlem failed, as no trade was concluded, as the prices were just too high for anyone to buy. Panic sparked quickly, prices collapsed within days, falling over 90%, and the market evaporated.

Those events had relatively inconsequential effects, as the defaults were limited to a few individuals. The settlements were agreed on locally, often through informal negotiations, and the broader financial system remained largely unaffected.

Tulipmania remains a classic case of speculative excess driven by leveraged, illiquid instruments, and a breakdown in correlation between the asset price and real economic value.

²Tulpen: (dutch) Plural form of "tulp", meaning "tulip" — a type of flowering plant known for its colorful, cup-shaped blooms

³Equivalent of the French Louis d'Or

⁴Called "wind-trade" because those contracts were based on future production, and not on available quantities

⁵A clearinghouse is a financial intermediary that facilitates the settlement of trades, ensuring that buyers receive their purchased assets and sellers receive payment. It reduces counterparty risk by guaranteeing the transaction even if one party defaults.

⁶A margin call occurs when the value of an investor's account falls below the required minimum margin. The investor must deposit additional funds or securities to the broker or clearinghouse to maintain their positions and cover potential losses.

⁷In modern futures contracts, the initial deposit ranges from 3 to 12%, plus the eventual margin calls

Wall Street Crash - 1929

The crash of 1929 occurred in the US and marked the end of the "Roaring Twenties", a decade of strong economic growth and booming equity markets. Fueled by widespread optimism, stock prices on the NYSE soared throughout the late 1920s.

By September 1929, the Dow Jones Industrial Average had more than tripled compared to its level four years prior.

A key driver of the bubble was margin trading, investors borrowed money to buy stocks, sometimes with initial margins as low as 10%, allowing them to take highly leveraged positions. Brokers extended easy credit, and millions of retail investors joined the market, many with little understanding of the risks.

Speculation intensified, detaching asset prices from underlying corporate earnings. Warnings from economists and the Fed about overvaluation were largely ignored.

The crash began on Thursday, October 24, 1929, "Black Thursday", when panic selling set in. This was followed by even more severe declines on Monday and Tuesday.

During those few days, the major stock indices lost more than 25% of their value.

The collapse continued into mid-November and the Dow Jones eventually fell nearly 90% from peak to trough over the following three years.

Banks, heavily exposed to the stock market and facing liquidity crises, began to fail. With no deposit insurance in place, consumer confidence plummeted, leading to bank runs, credit contraction, and a severe downturn in consumption and investment.

Unlike the Tulipmania episode, the 1929 crash had profound and long-lasting consequences. The financial sector imploded and the real economy followed. The US GDP shrank by nearly 30%, and unemployment soared above 20%.

The crash triggered the Great Depression, a global economic crisis that lasted a decade. In response, major regulatory reforms were introduced, including the Securities Act of 1933, the creation of the SEC⁸ in 1934, and the introduction of tighter banking regulations via the Glass-Steagall Act.

The 1929 crash remains one of the most severe financial collapses in history, demonstrating how excessive leverage, herd behavior, and regulatory gaps can amplify systemic risk and transform a market correction into a global depression.

"Black Monday" - 1987

The 1987 stock market crash, known as "Black Monday", occurred on October 19 1987, and remains one of the largest single-day percentage declines in history. The Dow Jones plunged about 22.6% in a single session, wiping out billions of dollars in market value.

Leading up to the crash, markets had experienced a strong bullish⁹ trend,

⁸Securities and Exchange Commission

⁹Rising in prices and investor optimism, in opposition to 'bearish' market

fueled by robust economic growth and increased portfolio investment globally. However, rising interest rates, growing trade deficits, and concerns about inflation started to create unease among investors.

A key feature of the 1987 crash was the widespread use of portfolio insurance strategies. These strategies used dynamic hedging¹⁰, often through index futures contracts, designed to limit losses by automatically selling stock index futures as markets declined. However, this selling accelerated downward pressure, creating a feedback loop of falling prices and increased selling volume. As more investors' portfolio insurance triggered sales, the market decline became self-reinforcing. In essence, the very mechanism intended to reduce individual losses ended up amplifying the overall market crash.

The crash was exacerbated by computer-driven trading programs and a lack of liquidity, causing markets to spiral down rapidly. Unlike previous crashes, the 1987 event was not triggered by a clear economic or political shock but rather by market dynamics and structural factors.

Despite the severity of the crash, the financial system did not collapse, and the economy remained relatively stable. The Federal Reserve and other central banks acted swiftly to provide liquidity and reassurance, preventing a broader financial crisis.

The 1987 crash highlighted the risks of automated trading and leveraged strategies, leading to regulatory changes including improvements in market circuit breakers: rules that temporarily halt trading if the market drops by a certain percentage within a short period as well as trading halts that pause individual stocks during extreme volatility.

These measures help prevent panic selling by giving investors time to re-evaluate information and allow liquidity to stabilize, improving overall market resilience. *Black Monday, 1987*, remains a powerful example of how market structure, investor behavior, and technological factors can combine to produce sudden and severe financial shocks, even in the absence of fundamental economic deterioration.

The US mortgage crisis - 2008

The 2008 crisis began with problems in the US housing market. Many home-buyers took out mortgages, loans to purchase property, but a large portion were 'subprime' mortgages, meaning they were given to borrowers with poor credit histories, sometimes unemployed, meaning they had a significantly higher risk of default.

These loans often had adjustable interest rates that started low but reset to much higher payments after a few years.

These rates could change because adjustable-rate mortgages were tied to market interest rates, so after the initial low "teaser" period, the rate would reset according to an underlying financial index plus a fixed margin, often resulting in higher monthly payments.

¹⁰Transaction(s) made to offset potential losses by reducing exposure to risk

To spread and sell the risk, banks pooled these mortgages into complex financial products called MBS, mortgage-backed securities, which investors could buy. Further complexity was added throughout CDO – collateralized debt obligations – which bundled various MBS and other assets into new securities with different risk levels. Another key instrument was the CDS – credit default swap – a form of insurance allowing investors to hedge or speculate on the risk of default on these securities without necessarily owning the underlying assets. A web of interdependent financial obligations with insufficient transparency was hiding huge risks and exposures to the housing markets.

Once the rates started going up, and defaults were limited but on the rise, the credit market did not react immediately, because banks were still solvent. At a certain point, when US housing prices began to fall, many homeowners defaulted on their loans, causing the value of MBS and CDO to decrease. (Try figuring out why!) Housing prices started falling because the increase in foreclosures added many houses to the market, reducing demand relative to supply.

Financial institutions that held large amounts of these securities suffered severe losses. This led to a liquidity crisis, banks became reluctant to lend to each other, fearing insolvency, resulting in frozen credit markets.

Major firms like Lehman Brothers declared bankruptcy, while others such as AIG required massive government bailouts to avoid collapse. In particular, Bear Stearns was acquired by JPMorgan Chase in March 2008 at a distressed price to prevent its sudden failure from destabilizing the financial system.

To stabilize the system, central banks and governments launched emergency programs, including the TARP¹¹, injected capital into banks, and lowered interest near zero percent.

In the aftermath, regulatory reforms aimed to prevent a repeat crisis. The Dodd-Frank Act introduced new rules, stricter capital and liquidity requirements to ensure banks hold enough reserves, mandatory stress tests simulating crisis scenarios, the Volcker Rule limiting risky proprietary trading by banks, and requirements for standardized derivatives trading through central clearing-houses to increase transparency and reduce counterparty risk.

¹¹Troubled Assets Relief Program

Quiz (1) - Major crashes

Each question has one or many possible answers.

1. Which social class was most affected by the collapse of Tulipmania prices?
 - ☐ Wealthy merchants
 - ☐ Middle-class artisans
 - ☐ Peasants
2. Before the 1637 crash, how were tulip contracts usually settled?
 - ☐ Through centralized exchanges
 - ☐ Through informal local agreements
 - ☐ By mandatory delivery of bulbs
3. Which factors contributed to the unsustainability of tulip contracts?
 - ☐ Lack of collateral
 - ☐ No legal enforceability
 - ☐ Speculation without real demand
4. Many subprime mortgages had interest rates that:
 - ☐ Stayed fixed for the life of the loan
 - ☐ Started low and then reset higher
 - ☐ Declined over time
5. Mortgage-backed securities (MBS) are:
 - ☐ Bonds backed by pools of mortgages
 - ☐ Insurance products for housing
 - ☐ Government subsidies
6. Which institutions failed during the 2008 crisis?
 - ☐ Lehman Brothers
 - ☐ AIG
 - ☐ Bear Stearns
7. During the early stages of mortgage crisis, the initial impact of widespread mortgage defaults was:
 - ☐ Housing prices spiked
 - ☐ Rising misconduct and early signs of stress in structured credit markets
 - ☐ Credit markets remained stable
8. Before the 2008 crisis, credit rating agencies often rated MBS and CDO as:
 - ☐ High risk
 - ☐ Investment grade
 - ☐ Junk bonds
9. A key structural problem in the US housing market before 2008 was:
 - ☐ Overproduction of housing in rural areas
 - ☐ Misalignment between borrower creditworthiness and loan terms
 - ☐ Excessive government subventions
10. In "The Big Short," which character is played by Brad Pitt?
 - ☐ Mark Baum
 - ☐ Ben Rickert
 - ☐ Jared Vennett

1.2 Basic financial instruments

Before delving into the technical mechanics of these instruments, it is essential to understand their fundamental roles within the financial system and the core features that define them.

Each instrument serves a specific purpose, whether it is to raise capital, transfer risk, gain exposure to a market, or optimize portfolio allocation. In the sections that follow, we will outline the structure, function, and key characteristics of major financial products such as stocks, bonds, derivatives, exchange-traded funds (ETFs), and indexes, laying the groundwork for a more detailed analysis later on.

Stocks

Also called equities, they represent ownership in a corporation. When an individual purchases a stock, they acquire a fractional claim on the company's assets, earnings, and, in many cases, voting rights in corporate decisions.

This ownership structure means that shareholders benefit when the company performs well, typically through potential capital gains or dividends, but also bear the downside risk if it performs poorly.

Companies issue shares to raise capital for business expansion, acquisitions, or debt reduction. This process occurs in the primary market, typically through an IPO – initial public offering – where the firm sells new shares to the public for the first time.

Once listed, stocks are freely traded on secondary markets such as the NYSE, NASDAQ, Euronext, or LSE. These exchanges provide liquidity and price discovery, allowing investors to buy and sell shares in real time.

Stock prices are influenced by various factors, including company fundamentals, (revenues, profits, guidance) macroeconomic conditions (interest rates, inflation, GDP), and investor sentiment.

Market participants range from retail investors to large institutional investors like pension funds and mutual funds.

Because stockholders are exposed to both market (systematic) risk and firm-specific (idiosyncratic) risk, stocks are considered riskier than bonds, but they typically offer higher potential long-term returns. Investors use a variety of strategies, such as diversification, technical analysis, or fundamental valuation, to manage their exposure and generate alpha¹².

Bonds

They are "fixed-income" instruments that represent a loan made by an investor to a borrower, typically a corporation, government, or public institution.

¹²In finance, alpha measures the excess return of an investment relative to a benchmark. It represents the portion of return attributable to the performance of the portfolio manager or the investment strategy, rather than market movements

When purchasing a bond, the investor becomes a creditor, not an owner, and receives regular interest payments called coupons over a defined period, followed by the repayment of the principal called face value, at *maturity* – the *expiry date* of the contract, also called *settlement date*.

Issuing bonds allows entities to raise capital without giving up ownership, as with stocks. The bond issuance occurs in the primary market, often through public offerings or private placements arranged by investment banks or underwriters.

Once issued, bonds can be traded in the secondary market¹³.

There are various types of bonds: sovereign bonds (issued by national governments, e.g. US Treasuries), municipal bonds (issued by local governments), and corporate bonds (issued by companies). Bonds differ by their expiry date, credit risk, coupon structure (fixed, floating, or zero), and currency denomination.

Credit rating agencies like Moody's, S&P, Fitch, assess a bond issuer's ability to repay, influencing the bond's interest rate.

Investment-grade bonds are considered safer, while high-yield often called "junk" bonds, offer higher returns to compensate for higher default risk.

Bond prices fluctuate in the secondary market due to changes in interest rates, inflation expectations, credit ratings, and liquidity conditions.

Because bond prices move inversely to interest rates, they are sensitive to central bank policy and broader macroeconomic developments.

Bonds are generally considered less risky than stocks, particularly government bonds from stable economies. They play a crucial role in portfolio diversification, capital preservation, and income generation, especially for risk-averse investors. However, they carry risks such as interest rate risk, credit risk, liquidity risk, and reinvestment risk.

Institutional investors such as pension funds, insurance companies, and banks are major participants in the bond market, often managing large fixed-income portfolios to meet long-term liabilities.

Derivatives

Derivatives are financial instruments whose value depends on the performance of an underlying asset, index, interest rate, or event. Common underlying assets include stocks, bonds, commodities, currencies, or benchmarks like interest rate curves or equity indexes.

Derivatives are primarily used for three purposes: hedging risk, taking spec-

¹³A secondary market is a financial market where existing securities are bought and sold between investors, rather than issued for the first time

ulative positions, or exploiting arbitrage¹⁴ opportunities.

There are four main types of derivatives: forwards, futures, options, and swaps.

Forwards and futures are agreements to buy or sell an asset at a predetermined price at a future date. Futures are standardized contracts traded on exchanges, while forwards are private agreements traded over-the-counter (OTC), which makes them more customizable but also more exposed to counterparty¹⁵ risk.

Vanilla options give the buyer the right—but not the obligation to buy (call option) or sell (put option) an underlying asset at a specified price before or on a set expiration date called the maturity. The seller of the option, however, must fulfill the contract if the buyer chooses to exercise it. Buyers pay an upfront (sometimes denoted "UF") premium for this right.

Swaps, on the other hand, are contracts in which two parties exchange cash flows. It's commonly used to manage interest rate or currency risk. For example, a company with a floating-rate loan might swap it for a fixed-rate one.

Derivatives can be used for risk management, for example, an airline might hedge against rising fuel costs using oil futures or options. But they can also be used to gain leveraged exposure to markets without owning the underlying asset. This makes them powerful but potentially dangerous tools.

Nowadays, derivatives are very regulated. As seen previously, after the 2008 financial crisis, where excessive exposure to complex derivatives like MBS, CDS and CDO played a major role in the bankruptcy of major US banks, almost collapsing the world economy, regulators introduced rules requiring many standardized derivatives to be cleared through central counterparties like clearing houses, to reduce opacity and counterparty risk.

Exchange-Traded Funds

ETFs are investment vehicles that pool together money from multiple investors to purchase a diversified portfolio of assets, such as stocks, bonds, or commodities, which are then traded on stock exchanges like individual shares.

Most ETFs are designed to track the performance of a specific index (e.g. the S&P 500 or Euro Stoxx 50), allowing investors to gain exposure to an entire market segment through a single, low-cost instrument. Some ETFs are passively managed, meaning they simply replicate an index, while others are

¹⁴In simple terms, an arbitrage is a way to make money without taking any risk. It involves a trade or strategy that cannot lose money and has a positive probability of generating a profit, essentially representing a "free lunch" in the market

¹⁵A counterparty is the other party in a financial transaction, such as a trade, contract, or loan. It is the person or institution you are dealing with and whose actions or ability to pay can affect the outcome of the transaction. Thus, counterparty risks are risks specifically related to the behaviour of counterparties. Ex: borrower's bankruptcy risk in a Bonds contract

actively managed and involve discretionary selection by portfolio managers.

Because ETFs are traded on exchanges, they offer the liquidity and flexibility of individual stocks: investors can buy and sell them throughout the trading day at market prices, use limit or stop orders, and even sell them short.

Today, ETFs span a wide range of strategies, from plain-vanilla index tracking to thematic, leveraged, inverse, or ESG-focused exposures, making them a core component in both retail and institutional portfolios.

Commodities

Commodities are raw materials or primary agricultural products that can be bought and sold, such as oil, gold, wheat, or copper. They are classified into two broad categories: hard commodities (natural resources like metals or energy) and soft commodities (agricultural products like coffee, sugar, or cocoa)

They are traded on specialized exchanges such as the CME (Chicago Mercantile Exchange), the ICE (Intercontinental Exchange) or the LME (London Metal Exchange), primarily through standardized futures contracts. These contracts specify the quantity, quality, and delivery date of the commodity, enabling participants to lock in prices and manage price volatility.

Market participants include producers, consumers, hedgers, and speculators. For instance, a chocolate manufacturer may hedge against rising cocoa prices by taking a short position in cocoa futures. If prices rise, the loss on the physical purchase is offset by the gain on the futures contract, allowing the company to secure predictable input costs. This kind of risk management will be thoroughly covered in a later section.

Commodities are influenced by a wide array of factors: global supply and demand dynamics, weather patterns, seasonal harvest cycles, geopolitical tensions, logistical bottlenecks, and currency movements, especially since most commodities are priced in US dollars.

Commodities are often used as a hedge against inflation, since their prices tend to rise when the purchasing power of money falls. They also provide portfolio diversification, due to their relatively low correlation with traditional asset classes like stocks and bonds.

Quiz (2) - Financial Products

Each question has one or many possible answers.

1. Stocks represent:
 - ☐ Ownership in a corporation
 - ☐ A fixed loan to a company
 - ☐ A derivative contract
2. Bonds are primarily used to:
 - ☐ Gain voting rights in a company
 - ☐ Lend money to an issuer in exchange for interest payments
 - ☐ Hedge against market volatility
3. A derivative's value depends on:
 - ☐ Its own intrinsic worth
 - ☐ The performance of an underlying asset or index
 - ☐ Payments issued by governments in example for capital
4. ETFs are designed to:
 - ☐ Track the performance of a portfolio of assets
 - ☐ Give loans to corporations
 - ☐ Be traded only once per day
5. Which type of bond generally offers the highest yield?
 - ☐ Sovereign bonds
 - ☐ Investment-grade corporate bonds
 - ☐ Junk bonds
6. Which derivative allows the buyer the right, but not the obligation, to buy or sell an asset?
 - ☐ Forward contract
 - ☐ Option
 - ☐ Swap
7. Which market is used for trading previously issued stocks and bonds?
 - ☐ Primary market
 - ☐ Secondary market
 - ☐ Derivatives market
8. Commodities can be classified as:
 - ☐ Hard or soft
 - ☐ Stocks and bonds
 - ☐ Derivatives and ETFs
9. Which of the following are common uses of derivatives?
 - ☐ Hedging risk
 - ☐ Speculation
 - ☐ Arbitrage opportunities
10. Bond prices move inversely to:
 - ☐ Interest rates
 - ☐ Stock prices
 - ☐ Donald's total Bitcoin position

1.3 Market participants

Financial markets bring together a diverse array of participants, each playing a unique role in ensuring market efficiency, liquidity, and risk management. Understanding who these participants are and how they operate is fundamental to grasping the dynamics of financial markets.

Retail Investors –

These are individual investors who buy and sell securities for their personal accounts.

Retail investors range from casual traders using online platforms to high-net-worth individuals managing complex portfolios.

While each retail investor generally operates on a smaller scale compared to institutions, their collective activity contributes significantly to market liquidity and price formation. Retail investors typically rely on brokers or digital trading platforms to access markets and often base decisions on financial news, research, or personal strategies.

For instance, in 2024, the SIFMA¹⁶ estimated that 18% of the US equity trading volume can be attributed to retail investors.

Institutional Investors –

Institutional investors manage large pools of capital on behalf of clients or beneficiaries. They include pension funds, mutual funds, insurance companies, endowments, and sovereign wealth funds.

These entities wield significant influence due to their substantial trading volumes and long-term investment horizons. Institutional investors engage in various strategies, from passive index tracking to active management, and often employ professional analysts and portfolio managers.

Their presence tends to stabilize markets, as they focus on fundamentals and diversification, provide consistent liquidity, and are less likely to react impulsively to short-term price movements. By managing large pools of capital with long-term horizons, spreading risk across many assets, and participating steadily in the market, institutional investors help reduce volatility, ensure smoother transactions, and contribute to more orderly market functioning.

Investment Banks and Broker-Dealers –

Investment banks facilitate capital raising, mergers and acquisitions, and market making. As intermediaries, they underwrite new securities issued in the primary market and provide liquidity in secondary markets by acting as market makers, buying and selling securities from their own inventory to facilitate smooth trading.

¹⁶Securities Industry and Financial Markets Association

Broker-dealers serve retail and institutional clients by executing orders and providing investment advice. They also operate trading desks, and historically some engaged in proprietary trading—trading with the firm’s own capital to profit from market movements. Today, regulations such as the Volcker Rule¹⁷ have restricted proprietary trading for many institutions, so this activity is less common in modern broker-dealers.

Given their central role in financial markets and potential for systemic impact, investment banks and broker-dealers are subject to strict regulatory oversight.

In the US, they must comply with rules set by the SEC and the FINRA¹⁸, including capital adequacy requirements, reporting obligations, and restrictions on conflicts of interest.

In Europe, regulations such as the Market Abuse Directive, Markets in Financial Instruments Directive, impose transparency and conduct rules. These measures aim to ensure market integrity, protect clients, and mitigate systemic risks stemming from excessive leverage or risk-taking activities.

Hedge Funds and Proprietary Trading Firms –

Hedge funds are privately managed investment pools that use diverse strategies such as long-short equity, arbitrage, derivatives trading, and leverage to achieve high returns. Some of the most successful ones use very advanced quantitative techniques, gain access to huge amounts of data, having virtually no budget limit to find and exploit trading strategies.

They often take on higher risks and can be more agile than traditional institutional investors, due to the fact that they face less strict regulations of their activity.

Portfolio managers play a key role in that kind of organization, as they are responsible for selecting, managing, and rebalancing investment portfolios in line with a fund’s strategy, risk tolerance, and performance targets.

They analyze macroeconomic trends, company fundamentals, and market dynamics to make asset allocation decisions, aiming to deliver returns above benchmark indices, generating α .

Still, it is unclear whether or not financial analysts and hedge fund managers can actually bring consistent insights on where markets are headed, as numerous studies¹⁹ have shown that outperformances are often not persistent over time.

¹⁷The Volcker Rule, part of the US Dodd-Frank Act of 2010, restricts banks from engaging in proprietary trading and limits their ownership of hedge funds and private equity funds. Its goal is to reduce risky trading activities that could threaten the stability of the financial system

¹⁸Financial Industry Regulatory Authority

¹⁹Notably, some sources explain the psychological biases and limits of investors and hedge fund managers, shedding the light on their inability to predict the market consistently: *Psychologie de l’investisseur et des marchés financiers*, Mickaël Mangot – 2005

Market Makers –

Market makers are liquidity providers who commit to posting continuous two-way quotations (bid and ask prices) for specific securities during trading hours. This quoting obligation ensures that counterparties can transact at any moment, minimizing execution delays and reducing market impact.

Their operations are especially critical in fragmented or less liquid markets, where order flow is sparse and price discovery is less efficient. By narrowing bid-ask spreads, they reduce implicit transaction costs for all market participants. Moreover, their quoting behavior is often algorithmically optimized to reflect changes in volatility, order book depth, other participant's behaviors and latency arbitrage risks, and is governed by sophisticated risk management frameworks to hedge inventory exposure.

In many jurisdictions and asset classes, designated market makers or primary dealers are incentivized or required by exchanges and regulators to fulfill minimum quoting obligations, in terms of time, size, and spread, to ensure orderly markets, particularly during stress events or for securities with low natural liquidity.

The Forex²⁰ market is often cited as the most efficient financial market, with over \$7.5 trillion traded daily and bid-ask spreads as tight as 0.1 pip²¹ for major currency pairs like EUR/USD. This immense volume stems from its central role in global trade, investment flows, and cross-border financial transactions, making currency exchange a necessity for governments, corporations, and investors worldwide.

This efficiency is largely attributable to institutional and high-frequency market makers that continuously provide deep liquidity across trading venues.

Their quoting algorithms dynamically adjust to market conditions, volatility, depth, and latency risk—enabling low transaction costs, minimal market impact, and near-frictionless execution. As a result, the Forex market exhibits rapid price discovery, intense competition, and negligible arbitrage opportunities despite its decentralized structure.

High-Frequency Traders –

High-frequency traders are specialized market participants who deploy advanced algorithms and low-latency infrastructure to execute a large number of trades at extremely high speeds, often holding positions for milliseconds or less, with order execution times measured in microseconds or even nanoseconds.

Their strategies include market making, arbitrage, and latency-based trading, all aiming to exploit short-lived inefficiencies across markets or instruments.

²⁰The foreign exchange market is the global decentralized marketplace where currencies are traded, operating 24/5 across all time zones

²¹A pip (percentage in point) is the smallest standard unit of movement in a currency pair's exchange rate, typically equal to 0.0001 for most major pairs

To gain speed advantages, HFT firms invest heavily in co-location services, meaning physically placing their servers as close as possible to exchange matching engines within the same data centers, to minimize signal travel time. They also utilize microwave transmission and other ultra-low-latency communication technologies to shave off microseconds in data transfer, enabling them to react faster than competitors to market events.

While HFTs are often criticized for potentially exacerbating volatility or causing self-reinforcing feedback loops, especially during market disruptions, they also play a key role in improving liquidity and tightening bid-ask spreads, particularly in highly fragmented electronic markets.

Their continuous presence in the order book enhances price discovery and facilitates trade execution for other participants. Regulators closely monitor HFT activity, requiring firms to implement safeguards such as kill switches, order-to-trade ratio limits, and pre-trade risk checks to mitigate systemic risks.

The relentless pursuit of trading speed contributes meaningfully to a global data center energy consumption, raising sustainability concerns²² in light of growing climate-related regulatory agendas.

Corporations and Governments –

Corporations and governments are foundational participants in financial markets, primarily as issuers of financial instruments.

Corporations issue stocks to raise permanent capital from investors and debt (bonds, commercial paper) to finance operations, acquisitions, or capital expenditures. These instruments are typically sold in primary markets via public offerings or private placements and subsequently traded in secondary markets, where their pricing reflects investor expectations regarding credit risk, profitability, and macroeconomic conditions. Large multinational firms also engage in treasury operations, actively managing liquidity, funding costs, and currency exposures across global markets.

Governments, through entities like national treasuries or debt management offices, issue sovereign debt to finance budget deficits, stabilize monetary conditions, or support long-term investment in infrastructure and public services. Instruments like US Treasury securities are widely considered risk-free benchmarks, serving as references for pricing across the yield curve and for discounting future cash flows in asset valuation.

Moreover, their disclosures and credit ratings often have ripple effects across markets, especially during times of financial stress, where a downgrade or policy announcement can trigger widespread repricing²³.

²²Impact of Data Centers on Power Consumption, Climate Change, and Sustainability - Bangalore university, India : https://www.researchgate.net/publication/378597789_Impact_of_Data_Centers_on_Power_Consumption_Climate_Change_and_Sustainability

²³The macroeconomic impact of news about policy and news about the economy in ECB announcements: https://www.ecb.europa.eu/press/research-publications/resbull/2018/html/ecb.rb180921.en.html?utm_source=chatgpt.com

Central Banks and Regulators –

Central banks, such as the ECB – European Central Bank, or the Fed – Federal Reserve, are essential actors in financial markets and the broader economy. They set key interest rates, like the ECB’s deposit facility rate, that influences the cost of borrowing for households, corporations, and governments.

By lowering rates, central banks aim to stimulate credit, investment, and consumption, thereby supporting economic growth. Conversely, raising rates is intended to curb inflation and moderate overheating.

A prominent example of central bank market intervention is quantitative easing. Between 2015 and 2022, the ECB conducted large-scale asset purchases—mostly sovereign bonds, under its Asset Purchase Programme and Pandemic Emergency Purchase Programme.

These programs injected liquidity into the financial system, compressed bond yields, and lowered borrowing costs across the euro area. In doing so, the ECB helped stabilize markets, support bank lending, and mitigate deflationary pressures during periods of weak growth.

Regulators such as the ESMA (European Securities and Markets Authority) or the SEC (Securities and Exchange commissions complement this role by enforcing transparency, run risk controls, and investor protection rules. It is crucial that regulators remain perfectly independent of the structures they oversee, although in practice this independence is not always fully achieved, as close ties between the financial industry and its regulators can sometimes lead to conflicts of interest or regulatory capture.

Regulatory frameworks impose disclosure obligations, trading rules, and capital requirements that shape how financial institutions operate.

These measures are designed to reduce systemic risk and ensure orderly, fair, and efficient markets. Market efficiency is a concept we will discuss throughout the next section.

Together, central banks and regulators form the institutional backbone of modern financial systems.

1.4 Mechanisms of the financial industry

Beyond capital allocation, the financial industry performs a number of essential functions that underpin modern economies.

Key mechanisms include the management and transfer of risk, the provision of liquidity, the promotion of market efficiency, and the aggregation and processing of information.

They ensure that financial markets operate smoothly, that prices reflect available information, and that firms and individuals can make informed decisions. By facilitating price discovery and enabling economic agents to hedge or transfer risk, the financial system contributes to financial stability and supports long-term economic growth.

1.4.1 Risk management & transfer

Risk management is a core function of the financial industry, allowing economic agents to mitigate adverse outcomes associated with uncertainty.

Insurance markets enable individuals and corporations to transfer specific risks, such as property damage, liability, or mortality, to specialized entities in exchange for a premium (a price), thereby stabilizing cash flows and protecting balance sheets.

Derivatives, including options, futures, and swaps, are powerful tools for hedging market exposures. Let us discuss two concrete examples of risk management using such products.

Example 1: Lehman Airways is calling...

Lehman Airways, an American airline, is heavily exposed to fluctuations in jet fuel prices, as fuel typically represents one of the largest operating costs for airlines, and even small fluctuations in jet fuel prices can significantly impact the company's profitability.

Next month, analysts expect possible price increases due to geopolitical tensions, the company's treasury department decides to hedge the next month's consumption of fuel costs using derivatives.

The airline expects to consume 2 million gallons²⁴ (7.57 million liters) of fuel next month, the period they want to hedge.

Thus, other this month's period, flights will consume approximately **47600** barrels of aviation gasoline.

This commodity is trading at \$90 per barrel.²⁵

In this scenario, a 10% increase on the barrel price yields an extra cost of

$$47600 * (1.1 * 90 - 90) = 428.400\$$$

Before finding a way to mitigate the risk of fuel price increase, let's give a little more details about the financial contract broadly used in such contexts : the option.

A "vanilla" option is a derivatives contract that gives the right (not the obligation) to buy or sell an "underlying" financial asset at a price and date, called the maturity or expiry, decided in advance.

An option that allows its owner to *buy* an asset is called a *call*. The *put* option lets the owner *sell* the underlying.

²⁴A gallon is a unit of volume commonly used to measure liquid fuels like gasoline, diesel, or jet fuel. 1 gallon = 3.785 liters

²⁵One barrel contains roughly 159 liters

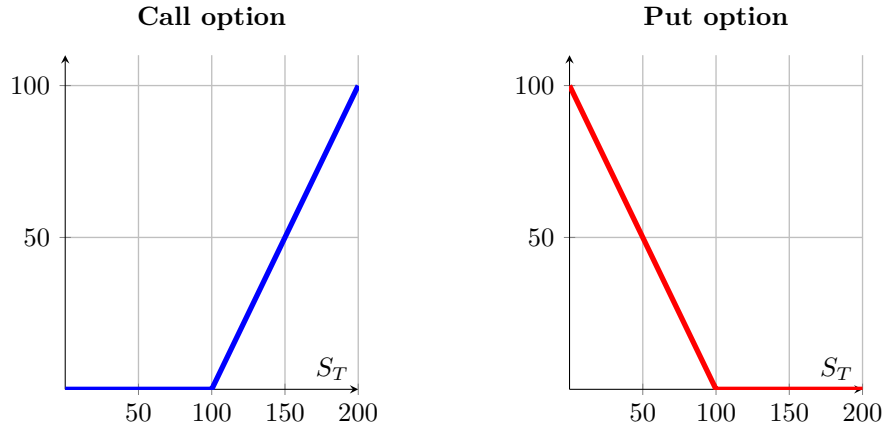


Figure 1: Payoff profiles of a European call and put option with strike $K = 100$ (no premium)

Note that these are the payoffs when buying the call or put. In practice, they cost money of course, thus, the graph must be offset to account for the premium.

For instance, a call option of the *AAPL* stock, with a strike of 1\$ and one month maturity lets the owner buy the Apple stock next week at a price of 1\$ (pretty advantageous, that contract would be very costly)

Now that we introduced this "option" contract, try thinking of a strategy the commodities risk manager could advise to hedge (protect) the group against the fuel market risk.

To hedge this risk, Lehman Airways can buy a call option on the aviation gas barrel at-the-money²⁶, expiring next month, paying a premium of \$3 per barrel. Let's see what the profit and loss (P&L) generated by the option contract is at maturity.

Scenario	Maturity price	Realized fuel cost	Options profit and loss
+10%	\$99	4.284 M\$	\$9 - \$3 premium = \$6/option
+2%	\$91.8	4.284 M\$	\$1.8 - \$3 premium = -\$1.2/option
-2%	\$88.2	4.198 M\$	-\$3/option
-10%	\$81	3.855 M\$	-\$3/option

Table 1: Impact of a 1-month ATM call option on AVGAS barrel for Lehman Airways for several scenarios

Since the option is at-the-money, the purchase price (strike) of the underlying stated by this contract will be done at fuel spot price, here at \$90, not at the maturity's market price.

²⁶An option is *at-the-money* when its strike price is equal, or very close, to the current market price of the underlying asset.

This strategy ensures price protection against a significant oil price surge, while still allowing the company to benefit from price declines.

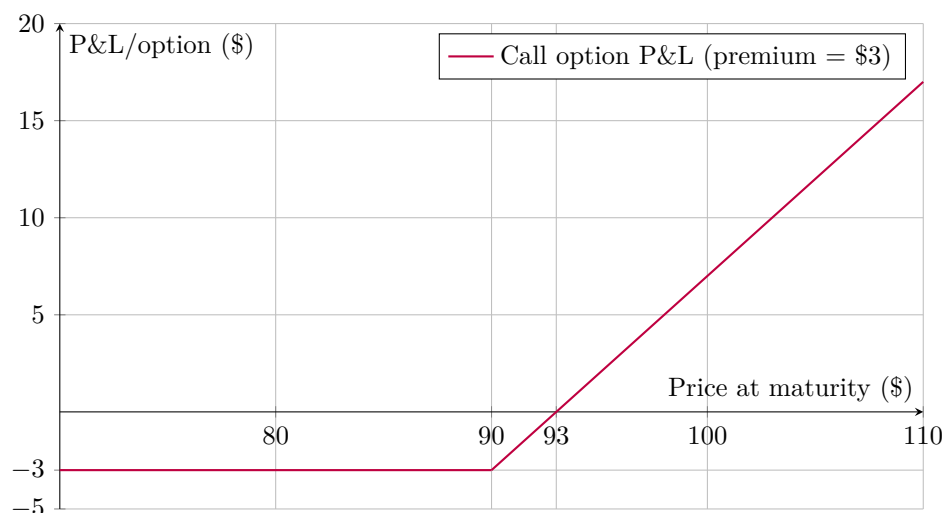


Figure 1: Net option P&L - The graph intersects $y = -3$ and $x = 93$.

This graph highlights that the call contract is nothing but an insurance covering the buying against underlying price increase, yielding positive P&L if the price at maturity has increased. (Depending on the premium and price increase magnitude. Note that we define the "break-even" maturity price as the price at which the premium is perfectly compensated by the contract's P&L, here, \$93)

At maturity, Lehmann Airways exercises the option if it's in their interest to do so.

Note that it is possible to exercise an option even though it would yield an immediate negative P&L. No one does though²⁷

Example 2: Bankruptcy Risk

Goldman Stanley (GS), a major international financial institution, holds a substantial position in corporate bonds issued by Pear Computers Inc. (Pear), a tech company.

These bonds are trading at \$1,000 each, with a 1-year maturity and a 1% annual coupon, which is the fixed interest payment the bondholder receives each year, expressed as a percentage of the bond's face value.

GS analysts are concerned that Pear's creditworthiness may deteriorate, following a recent decline in sales and an increase in debt levels. In particular, GS is highly exposed to the risk of Pear defaulting on its obligations. A default

²⁷Except in very specific scenarios, like: <https://www.investopedia.com/terms/a/americanoption.asp?>

occurs when a borrower fails to meet the legal terms of a debt agreement, typically by missing scheduled payments of interest or principal, or by breaching other contractual conditions.

Creditworthiness is the evaluation of a borrower's ability to repay debt and is crucial for lenders to determine the level of risk involved in lending money.

The assessment is based on several factors.

Credit history, which includes details on how well they've managed past debts, such as whether they made timely payments or had defaults or bankruptcies. A solid credit history suggests the borrower can responsibly manage future debt. Lenders also look at the borrower's income stability to ensure they can consistently meet repayment obligations. The more predictable and stable the income, the less risky the borrower is perceived to be.

Another key factor is the debt-to-income ratio, which compares the borrower's debt obligations to their income. A lower ratio means the borrower is less burdened by debt and more likely to handle additional financial commitments. Assets and savings are also considered, as they provide a cushion for repaying the loan in case of financial hardship.

For secured loans, lenders may require collateral, such as property or equipment, which is an asset pledged by the borrower to guarantee repayment. This reduces risk by providing something the lender can seize if the borrower defaults. A borrower with strong creditworthiness, indicated by a good credit score, stable income, low debt, and sufficient assets, is more likely to receive favorable loan terms, such as lower interest rates and higher borrowing limits. This evaluation is often performed by credit rating agencies such as Moody's or Standard & Poor's.

Back to Goldman Stanley's office where our counterparty risk analyst is concerned about Pear's risk of default. Try thinking of a strategy GS could use to cancel its exposure to Pear's default.

To mitigate this credit risk exposure, GS decides to hedge its position using a Credit Default Swap (CDS).

It involves two parties: GS (the protection buyer) and a Hedge Fund (the protection seller). The protection buyer will pay an annual premium to the protection seller in exchange for receiving compensation if a "credit event" occurs with Pear.

A *credit event* is a predefined circumstance in a CDS contract, such as bankruptcy, failure to pay, or debt restructuring, that triggers the protection seller's obligation to compensate the protection buyer for losses resulting from a borrower's inability to meet its debt obligations.

- **GS Exposure:** Ownership of \$10 million in bonds issued by Pear.
- **CDS Contract:**
 - **Notional:** \$10M (Total exposure amount)

- **Premium:** 2% annually on the notional amount
- **Credit event(s):** Default, bankruptcy, or significant restructuring of Pear
- **Maturity:** 1 year (matching the maturity of the bonds)

Table of Scenarios:

In this case, GS and the Hedge Fund agree to the following annual premium payments and various potential scenarios of credit events or no credit events: Note that in the P&L computation, we have to account for the coupon rate payment of the bonds occurring at maturity in the cases where Pear remains solvent.

Scenario	Pear situation	Bonds at maturity (M\$)	CDS P&L (M\$)
no event occurs	remains creditworthy	10.1	-0.2 (premium)
credit downgrade	rating drops, no default	10.1	-0.2 (no payout)
default	defaults	0	9.8 (payout)
restructuring	restructuration, no default	7	4.8 (partial payout, if any)

Total P&L (M\$)

Scenario	Without CDS	With CDS
no event occurs	10.1	9.8
credit downgrade	10.1	9.8
default	0	10
restructuring	7	12

Table 2: Impact of the CDS on GS credit risk exposure, comparison of total P&L with and without CDS protection

This strategy ensures that GS is protected against a total loss if Pear defaults on its bond payments, while the Hedge Fund earns a regular income from the premiums.

In our example, GS pays a 2% annual premium (which is \$200,000/year on the \$10 million notional) to the Hedge Fund in exchange for protection. If Pear defaults, the Hedge Fund pays the full notional value (\$10 million) to GS.

Let's summarize the financial positions for the Bank and the Hedge Fund in each scenario:

1. **No Default:** GS loses the annual premium of \$200,000, but the bond value remains intact at \$10 million. GS receives \$100,000 from Pear in bonds coupon. The Hedge Fund receives the annual premium, with no payout, realizing a positive P&L.
2. **Credit Downgrade:** The bond value drops to \$9 million due to the downgrade in the market, but no default occurs. GS loses the premium,

and the Hedge Fund receives it with no payout. GS still receives \$100.000 from Pear in bonds coupon, as the coupon is computed from the bonds notional, not the market value of the bond. Not that in this scenario, if no default occurs, it won't affect the final P&L of both counterparties at maturity. A downgrade characterizes a message from the market that the underlying entity emitting the bond is more likely to default.

3. **Default:** The bond value falls to \$0 in the event of default. The Bank receives \$10 million from the Hedge Fund as a full payout, as agreed in the CDS contract. Pear does not pay the bonds coupon to GS.
4. **Restructuring:** The bond value drops due to restructuring. In a CDS contract, a restructuring is a credit event where a borrower's debt terms are changed due to financial distress. For example, by reducing interest, extending maturity, or postponing payments. If the restructuring is deemed unfavorable to creditors, it may trigger a CDS payout. Not all restructurings qualify, only those defined by ISDA²⁸ rules apply. The Hedge Fund may offer a partial payout, covering part of the losses, but GS still loses the premium. In a restructuring case, the payment of the coupon bonds can be happening or not, depending on the CDS contract. In this case, let's say it does not.

This example demonstrates how a CDS contract can provide a hedge against severe credit events, such as default or restructuring, for financial institutions holding bonds in at-risk companies.

Anecdote: Michael Burry and the Invention of Subprime CDS

In *The Big Short*, Michael Burry (portrayed by Christian Bale) discovers around 2005 that the US housing market was built on highly unstable subprime loans. His key decision was to invest in credit default swaps on mortgage-backed securities, essentially betting on the collapse of these toxic loan structures.

At the time, however, CDS on subprime mortgage bonds did not exist. In 2005, Burry had to persuade Deutsche Bank, Merrill Lynch, Goldman Sachs, and Morgan Stanley to construct these instruments specifically for him.

For months, he paid premiums while being widely dismissed or ignored by the financial establishment, until the market crashed, and his fund roughly \$700M in profits.

Bonus trivia: In real life, Burry was known to listen to heavy metal music constantly while working. The scene in the film where Christian Bale's character is shown playing drums to heavy metal music is based on his real-life hobby.

This movie is highly recommended to both understand the mortgage crisis and to have a good laugh at some of the absurdities of the financial industry that existed before 2008.

²⁸International Swaps and Derivatives Association. It is the global trade organization that develops standardized documentation, definitions, and protocols for over-the-counter (OTC) derivatives, including CDS contracts

1.4.2 Market efficiency

The *CFA institute* defines this concept as: Efficient markets imply informative prices, that accurately reflect available public information about fundamental values.

Firstly, note the subjectivity of this definition.

In such a market, no investor can consistently achieve returns above the average market return by using public information, because prices already incorporate it.

This means securities are fairly priced at all times, making it difficult to find undervalued or overvalued assets based on public data.

It is the role of market regulators to continually enhance market efficiency. However, perfect efficiency has not yet been achieved – and likely never will be – meaning that real-world markets remain inherently *inefficient*. Let us explore the main characteristics of inefficient markets.

These include information asymmetry, delayed price adjustment, behavioral biases, market frictions and transaction costs, limits to arbitrage, structural and regulatory factors related to market design and rules affecting price discovery and liquidity issues.

1. Information Asymmetry

Information asymmetry occurs when one party in a transaction has more or better information than the other. This imbalance can significantly impact market efficiency, as it may lead to mispricing of assets and sub-optimal allocation of resources.

For example, sellers may have more knowledge about the quality of a product than buyers, or insiders may possess private information about a company's prospects that is not publicly available.

2. Delayed Price Adjustment

Delayed price adjustment refers to the phenomenon where market prices do not immediately or fully reflect new information. Various factors can cause this lag, including slow dissemination of information, behavioral biases among investors, or institutional constraints such as trading hours and liquidity limits.

As a result, prices may underreact or overreact initially, leading to temporary mispricing. This inefficiency creates opportunities for informed traders to earn abnormal profits by exploiting the delay before prices reach their fair value.

Over time, however, arbitrage and market forces tend to correct these inefficiencies, but delays can persist, affecting short-term market dynamics and volatility.

3. Behavioral Biases

Behavioral biases represent systematic deviations from rationality in investors' decisions, often caused by emotional influences, heuristics, and cognitive limitations.

The most important biases include overconfidence bias, where investors overestimate their knowledge or ability to predict market movements leading to excessive risk-taking.

Loss aversion, which means the pain of losses outweighs the pleasure of equivalent gains.

Herding behavior, where investors tend to follow the majority often resulting in asset bubbles or crashes.

Anchoring bias, involving reliance on initial information such as past prices that skews judgment.

Confirmation bias, the tendency to seek information that supports existing beliefs while ignoring contradictory evidence. One very well known context where this bias poses a significant risk, is in the process of *back-testing*²⁹ trading strategies.

4. Market Frictions

Market frictions refer to various real-world imperfections and costs that impede the smooth functioning of financial markets.

These include transaction costs such as bid-ask spreads, commissions, taxes, and delays in trade execution, all of which can significantly reduce the profitability of trading strategies that appear profitable in frictionless theoretical models.

Additionally, liquidity constraints and regulatory restrictions further limit the ability to implement trades at desired prices or volumes, making it essential to account for these factors in quantitative finance and strategy development.

For example, attempting to execute a very high volume order in a less liquid market can move prices against the trader, causing market impact costs that reduce expected profits. This price slippage occurs because the large order “offsets” available liquidity, forcing execution at less favorable prices.

“Market microstructure” has a whole dedicated branch within financial mathematics³⁰, partly focusing on modeling and quantifying the effects of large trades on asset prices.

²⁹Backtesting is the process of evaluating a trading strategy using historical market data to assess its potential performance. Due to confirmation bias, quantitative analysts, data analysts, traders may give undue weight to favorable backtest results that align with their expectations, while discounting or ignoring data that reveals flaws or poor performance, which can lead to overestimating the strategy's real-world effectiveness

³⁰Laruelle & Lehalle, *Market Microstructure in Practice* (2013) explores how financial markets operate at a detailed level, covering trading mechanisms, order types, liquidity, and strategies used by practitioners to optimize execution and manage market impact

1.4.3 Price discovery

Price discovery is the fundamental process through which the value of a financial asset is determined in the market. It results from the ongoing interaction between buyers and sellers as they exchange information, preferences, and expectations, ultimately arriving at a price that reflects the collective consensus. Understanding price discovery is essential to grasp how markets function and respond to new information.

Before the advent of electronic markets, price discovery took place on physical trading floors through a system known as *open outcry*.

Traders representing buyers and sellers would gather in pits, circular areas of the exchange, and shout, gesture, and use hand signals to communicate bids and offers in real time.



Figure 2: S&P 500 pit of the CME Group’s Chicago Board of Trade — 2012

This chaotic but structured environment allowed prices to form through direct human interaction and negotiation.

Floor brokers competed to match orders on behalf of clients, and market makers provided liquidity by quoting buy and sell prices. While effective, this system had limitations: it was less transparent, way slower, and more prone to human error or information asymmetry.

However, open outcry fostered a unique ecosystem where experienced traders relied heavily on intuition, voice inflection, and crowd psychology to gauge market sentiment and execute trades.

Nowadays, the London Metal Exchange (LME) is one of the last marketplaces that still maintain a small open-outcry trading pit for certain contracts, preserving a piece of this traditional price discovery method. This pit continues to operate a few days a week, serving as a symbolic and functional reminder of the historical roots of trading.



Figure 3: "The Ring" – London Metal Exchange, created in 1877, the world's largest market in forwards, futures & options on base metals

It wasn't until the 1990s and early 2000s that most exchanges transitioned to fully electronic systems, where matching engines and digital order books replaced human intermediaries in the price formation process. In the next section, we focus on the structure and function of order books, which form the backbone of electronic trading.

Order book –

An order book is an electronic list of buy and sell orders for a specific financial asset, organized by price level in real-time. It shows the quantities traders are willing to buy or sell at different prices, providing transparency, and is often accessible to market participants through real-time or slightly delayed data feeds, allowing them to observe supply and demand dynamics.

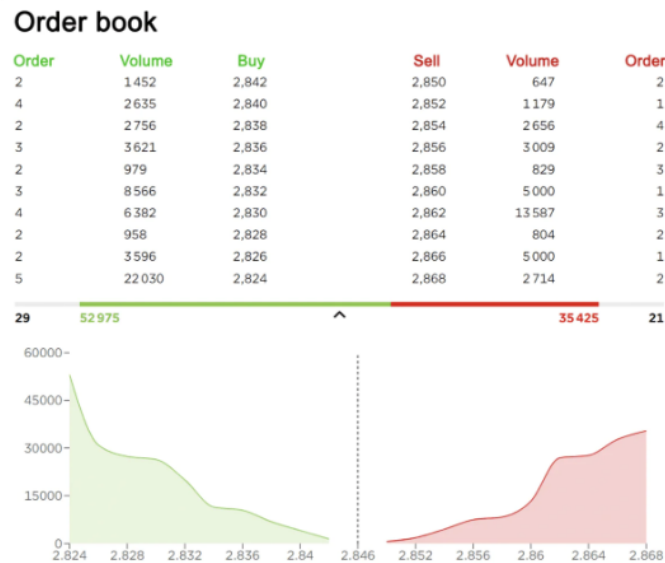


Figure 4: Order book representation

Here is a representation of an order book. The green zone represents the *bid* prices (buy side) associated to their respective volume on the y-axis. The red zone represents the selling orders at *ask* prices. As shown, the price forms in between.

What is the spread and why does it exist?

The ask is higher than the bid because buyers aim to purchase at the lowest possible price, while sellers seek to sell at the highest possible price. This difference, known as the *bid-ask spread*, compensates market makers for providing liquidity, reflects supply-demand imbalances, transaction costs and information asymmetry.

Let's imagine the following situation: You're working at an electronic market firm, and the matching engine suddenly breaks down. Until the issue is fixed, you must perform the order matching calculations manually, from all the data available in the order book. So, you receive orders for a given product: buy, sell orders, for different price and quantities, having different order types etc...

Consider how you would: (without stressing)

determine the market price, and match buy and sell orders to balance supply and demand and keep the market running smoothly (maybe with a little more latency?)

Matching engine functioning:

One important remark is that, as quite some mathematical problems, order matching can be solved in a lot a different ways.

The *FIFO* (first in, first out) algorithm, also called *Price/Time* matching algorithm is one of them.

Let's break down its implementation, considering the following fictive order book.

Side	Price	Quantity	Timestamp
Sell	104	100	09:30:30
Sell	105	200	09:30:01
Sell	105	150	09:30:15
Buy	103	100	09:29:45
Buy	102	200	09:29:50
Buy	102	250	09:29:55

Table 3: Order book at equilibrium: no possible matching

In this order book, separated into two “sub-order-books” containing only buy and sell orders, note that no order can be matched and executed. Indeed, the highest price someone is willing to pay to buy the security is lower than the lowest price someone is willing to accept to sell it. Thus, no transaction can occur.

Note that orders are ranked by price and timestamp, the less recent orders will be matched firstly.

Side	Price	Quantity	Timestamp
Sell	104	100	09:30:30
Sell	105	200	09:30:01
Sell	105	150	09:30:15
Buy	104	30	09:30:40
Buy	103	100	09:29:45
Buy	102	200	09:29:50
Buy	102	250	09:29:55

Table 4: A new buying order crossing the lowest ask was submitted

In this situation, two orders can be matched together. Indeed, the new buying order was submitted at a price superior or equal to the lowest selling price.

At \$104, a quantity of 100 appears on the sell side, against a quantity of 30 on the buy side. Thus, a quantity of 30 can be matched, partially executing the selling order, that will have a remaining volume of 70 to execute.

Side	Price	Quantity	Timestamp
Sell	104	70	09:30:30
Sell	105	200	09:30:01
Sell	105	150	09:30:15
Buy	103	100	09:29:45
Buy	102	200	09:29:50
Buy	102	250	09:29:55

Table 5: Order book after execution: trade at 104 for 30 units, partially filling the executed ask order

And again, we are in an equilibrium situation, where no orders can be matched together.

Note that the market price of the security dealt in this order book is the *last traded price*. In this order book, right after the last execution at \$104, the market price is updated to \$104.

Also note that the more trades happen, the more the market price is updated, reflecting the newest information and buyers/sellers sentiment, analysis, and incentives.

As said previously, it is important to note that the FIFO algorithm we presented is only one possible matching method; other algorithms, such as those based on Walrasian equilibrium, can match multiple orders simultaneously at a single price. Furthermore, this section does not cover the full range of order types and special instructions that exist in modern electronic markets.

Quiz (3) - Mechanisms of the Financial Industry

Each question has one or many possible answers.

1. Risk management in financial markets is primarily about:
 - ☐ Maximizing short-term profits
 - ☐ Transferring, mitigating, or diversifying risks
 - ☐ Avoiding all exposure to financial markets
2. Insurance markets allow economic agents to:
 - ☐ Transfer risk to insurers
 - ☐ Eliminate risk entirely
 - ☐ Guarantee profits on investments
3. In an efficient market, according to financial theory:
 - ☐ Prices fully reflect all available information
 - ☐ Prices move randomly without explanation
 - ☐ Investors can consistently beat the market using private information
4. Which of the following is NOT a source of market inefficiency?
 - ☐ Delayed reaction to new information
 - ☐ Mispricing of assets
 - ☐ Perfect information symmetry
5. In a limit order book, a persistent order imbalance (more aggressive buy orders than sell orders) typically leads to:
 - ☐ Upward pressure on prices as liquidity at the ask is consumed
 - ☐ No systematic effect, since order books are memoryless
 - ☐ A widening of the bid-ask spread without affecting mid-price
6. Market frictions such as transaction costs and liquidity constraints:
 - ☐ Increase market efficiency
 - ☐ Distort trading strategies and execution
 - ☐ Have no impact on financial markets
7. Price discovery is the process of:
 - ☐ Determining asset prices through supply and demand
 - ☐ Governments fixing interest rates
 - ☐ Rating agencies assigning credit ratings
8. The bid-ask spread in an order book:
 - ☐ Compensates market makers for providing liquidity
 - ☐ Represents a trading fee charged by the exchange
 - ☐ Eliminates supply-demand imbalances
9. Which of the following are common types of orders in an electronic order book?
 - ☐ Market orders
 - ☐ Limit orders
 - ☐ Stop-loss and take-profit orders
10. A trader holding a short call option position is exposed to:
 - ☐ Limited loss potential
 - ☐ Unlimited loss potential
 - ☐ Ending up in jail for insider trading and market manipulation

2 Python for Finance

Python is a high-level, interpreted programming language known for its simplicity, readability, and versatility. Created by the Dutch engineer *Guido van Rossum* and first released in 1991, Python emphasizes clear syntax and code readability, making it accessible for beginners while being powerful enough for experts.

Python has a large community that has tackled nearly every problem domain, providing libraries for tasks from data science and machine learning to web apps and automation. This rich ecosystem lets developers build solutions quickly without reinventing the wheel.

It remains widely used in the financial industry, where its ability to quickly process data, perform complex numerical computations, and interface with APIs makes it ideal for tasks such as algorithmic trading, risk management, pricing models, and portfolio optimization. In this chapter, we'll focus specifically on how Python is applied in finance and explore the key libraries and tools that make it so effective in this domain.

The main content of this chapter is hosted on GitHub, to allow code updates easily, and to let you run it, play with it, improve it, fail and learn.

2.1 Installations

To effectively use Python for finance, you first need to set up your Python environment and tools for interactive development.

Python can be installed in two main ways: either through the **Anaconda** distribution (recommended for beginners in finance and data science) or directly from the official Python website.

If you prefer a minimal installation, download Python from: <https://www.python.org/downloads/>

PyCharm is a professional-grade **Integrated Development Environment (IDE)** for Python. An IDE is a software application that brings together all the tools you need for programming, a code editor, a debugger, a project manager, and integration with version control systems, into one unified interface.

While Jupyter Notebooks are great for exploratory work and data analysis, an IDE like PyCharm is more suited for developing larger, real-world applications.

Download PyCharm from: <https://www.jetbrains.com/pycharm/download/> (the "Community" Edition is free) Once installed, you can create a new Python project, configure the Python interpreter, and start writing, running, and debugging scripts in a structured environment.

Jupyter Notebooks is one of the most popular ways to work with Python in finance and data science. These notebooks allow you to combine live code, equations, visualizations, and narrative text, making them ideal for exploration and presentation.

The **Anaconda** distribution is highly recommended. It bundles Python with Jupyter Notebooks. Anaconda comes with **conda**, a powerful package and environment manager.

You can download Anaconda from its official website: <https://www.anaconda.com/products/distribution>

Python packages are collections of modules and code bundled together to provide specific functionality or features, such as data analysis, visualization, or web requests. These packages let you reuse and share code easily without reinventing the wheel. **pip** is Python's standard package manager. It allows you to install additional Python packages from the Python Package Index (PyPI), a vast repository of reusable software components.

If you have installed Python without Anaconda, you can install Jupyter and other packages using **pip** commands:

```
pip install jupyter numpy pandas matplotlib
```

Similarly, to install any other package, use:

```
pip install package_name
```

For example, to install the HTTP library **requests**, type:

```
pip install requests
```

Once installed, launch a Jupyter Notebook by opening your terminal or command prompt and running:

```
jupyter notebook
```

This command opens a browser interface where you can create, edit, and execute Python notebooks interactively.

2.2 Python core

In this section, we will review intermediate Python concepts. This chapter is intended for readers who already have a little programming experience in Python, we assume you're beginning to be comfortable with the basics such as variables, loops, and functions, classes, object-oriented-programming. We will focus on concepts that are less trivial but highly relevant, especially in professional environments and technical interviews for Python or finance-related roles.

Find this section in the following **Jupyter Notebook** on the Github course repository : <https://github.com/SK8gh/EFREI/blob/main/Training/Python%20Core.ipynb>

2.3 Specific Python use-cases in finance

To demonstrate the power of Python, let's build together, throughout the following sections, some scripts that will perform data analysis of financial time series, as well as derivatives pricing scripts.

To do so, we will need to implement scripts that request data, process it, display it, and create indicators and analytics: <https://github.com/SK8gh/EFREI/tree/main/Bloom>

2.3.1 Libraries

The following libraries are mandatory when using Python for finance.

- **pandas** — a powerful library for data manipulation and analysis. In finance, pandas is widely used to work with time series, tabular data, and complex datasets.
Its **DataFrame** structure allows easy filtering, grouping, and transformation of large datasets.
Note that the dataframe operations are kind of like SQL actions – groupby, filtering, merge etc...
- **numpy** — a fundamental library for numerical computing in Python. It provides efficient multidimensional arrays and mathematical functions. It is used all the time for vectorized calculations and matrix operations.
- **matplotlib** — a comprehensive plotting library for creating static, animated, and interactive visualizations.
- **scipy** — Useful for advanced mathematical, statistical, and optimization tasks.

In the course repository: <https://github.com/SK8gh/EFREI/tree/main/Training>

You will find the following training files: `pandas.ipynb`, `numpy.ipynb` to learn the main objects and methods of those libraries.

As well as a test notebook file to assess your understanding of pandas, numpy, and core Python concepts: `Questions.ipynb`

2.3.2 Market data acquisition

Within organizations, market data (or other kinds of data) are requested by applications all the time. Requesting data can be done through a variety of channels depending on the architecture and available systems.

In many cases, Python scripts interact with internal data servers or APIs to pull real-time or historical prices, volatilities, interest rates, or economic indicators.

For example, large institutions may provide in-house REST or gRPC APIs, often wrapped in proprietary Python libraries to simplify access. In other contexts, the data may come from commercial vendors (e.g. Bloomberg, Refinitiv, ICE), where Python bindings such as `blpapi` or `refinitiv.data` allow querying market snapshots, streaming updates, or time series.

Beyond direct vendor APIs, Python is also used to query databases storing curated market datasets, often in combination with libraries like **pandas** for immediate in-memory processing. For high-frequency or event-driven use cases, websocket connections or message queues (e.g. Kafka, ZeroMQ) may be employed, allowing Python applications to react to incoming data with minimal latency.

Once retrieved, data typically undergoes a transformation layer in Python, cleaning, aligning timestamps, adjusting for corporate actions, before feeding into analytics, risk models, or trading algorithms. This integration of acquisition and preprocessing is one of Python's strengths in the finance domain.

Let's go through an example of data acquisition and processing.

```

OPEN_TIMESTAMP                2023-01-01 00:00:00
OPEN                          16541.77000000
HIGH                          16545.70000000
LOW                            16508.39000000
CLOSE                          16529.67000000
VOLUME                        4364.83570000
CLOSE_TIMESTAMP               2023-01-01 00:59:59.999000072
QUOTE_ASSET_VOLUME            72146293.58672290
TRADES_NUMBER                  149854
TAKER_BUY_BASE_ASSET_VOLUME    2179.94772000
TAKER_BUY_QUOTE_ASSET_VOLUME    36032352.87035720
IGNORE                         0
Name: 0, dtype: object

```

Figure 5: Bitcoin data requested using Python

Cryptocurrencies are digital assets secured by cryptography, enabling peer-to-peer transactions without intermediaries like banks.

Binance, founded in 2017, has grown into the world's largest cryptocurrency exchange by trading volume.

It provides a comprehensive API that allows developers and traders to access market data, manage accounts, and execute trades.

The Binance Python API allows developers to interact directly with Binance's trading platform without using the web interface. It provides access to market data such as current prices, historical candlesticks, and order book depth.

Users can also manage their trading accounts, including balances, open orders, and trade history. The API supports order placement, including market, limit, and stop orders, as well as order cancellation and modification.

For real-time needs, it delivers continuous, low-latency updates on market instruments, such as live ticker prices or changes in order book depth, which are essential for algorithmic trading systems requiring up-to-the-second information.

The official Python wrapper, `python-binance`, offers a high-level interface to these features. After installing it via `pip`, running the command

```
pip install python-binance
```

developers create a `Client` object by providing an API key and secret from their Binance account dashboard.

Once authenticated, methods return structured Python dictionaries with requested data. Placing an order can be done simply by calling the appropriate methods with the right parameters.

For live updates, the subscription to market data streams can be requested, triggering a callback function whenever new information arrives.

Binance enforces strict rate limits on API usage. Exceeding these limits can lead to temporary bans.

Let's now focus on the technical aspects of accessing and processing data through the API.

```
client = Client("API_KEY", "API_SECRET")
price = client.get_symbol_ticker(symbol="
    BTCUSDT")

print(price)
# {'symbol': 'BTCUSDT', 'price':
    '118566.01'}
```

This snippet requests the price of a given product traded in Binance, in this example BTCUSDT is the trading pair symbol used on Binance and many other cryptocurrency exchanges to represent the exchange rate between Bitcoin and Tether (USDT), a stablecoin tied to the US dollar.

Thus, BTCUSDT indicates how much \$ one Bitcoin is worth.

2.3.3 Financial time series analysis

In the code associated to this section, we will examine key metrics such as returns, volatility, and correlations between assets. We will analyze the distributions followed by returns, and create trading indicators and candlestick charts.

2.3.4 Data visualization

Again, in the Jupyter Notebooks associated to this section, we will discuss graphical tools to plot price series, distributions of returns, build candlestick charts, trading indicators and correlation matrices.

3 Mathematical modeling - Black Scholes

Mathematical modeling serves as a formalized and structured way to understand, quantify, and predict the behavior of complex systems.

It is simply everywhere: from predicting weather patterns and modeling the spread of diseases, to designing aircraft, simulating traffic flow in cities, forecasting energy demand, modeling population growth, and creating realistic animations in computer graphics.

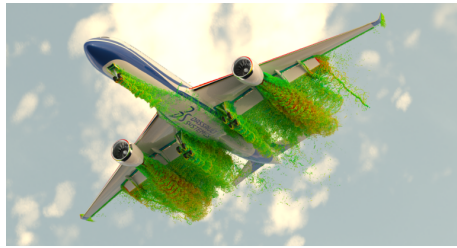


Figure 6: *Dassault Systèmes SIMULIA* fluid simulation solutions help aerospace engineers design and validate aircraft aerodynamic performance more efficiently, reducing reliance on costly physical testing while meeting strict regulatory requirements

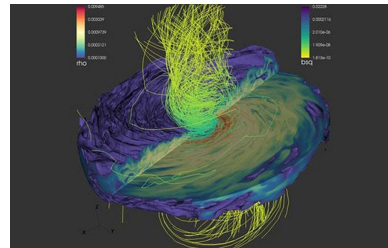


Figure 7: Supermassive black hole merge simulation, showing how gas disks produce distinct light signals that, alongside future gravitational wave detections, could help astronomers identify and study these cosmic collisions

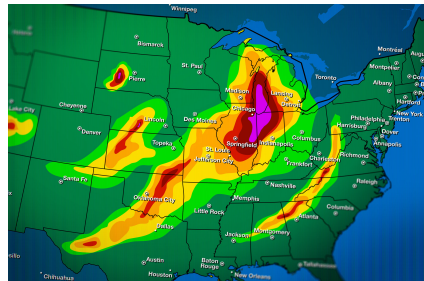


Figure 8: Weather simulations model atmospheric conditions to help scientists and meteorologists make more accurate forecasts, improving safety, planning, and disaster preparedness

Financial markets are inherently uncertain and influenced by countless variables. Prices fluctuate due to economic data, investor sentiment, geopolitical events, corporate earnings reports, the woke culture, natural disasters, technological innovations, changes in monetary policy, the POTUS tweets, etc...

Fundamentally, a *mathematical model* is an abstract description of a concrete system using mathematical concepts and equations.

By making assumptions and using mathematical and computer science tools, a framework can be defined to abstract the underlying systems and ultimately try to produce inferences and calibrations.

Calibrating a model means inputting real-world data into it to find a set of parameters that will best reproduce observed real-life outputs or behaviors, typically by applying iterative methods, minimizing the discrepancy between model outputs and actual data through advanced gradient descent techniques³¹.

Note that making assumptions is an essential part of creating a model, it is essential to recall that those assumptions may be inaccurate and not verified in reality.

Once a model that "works", meaning it allows us to make meaningful inferences about reality has been established, we can begin an iterative process of refining its assumptions. This may involve simplifying or complicating the model to evaluate whether the new versions offer better, worse, or simply different characteristics and predictive or explanatory capabilities compared to the original.

In this chapter, we will particularly investigate the Black–Scholes model (sometimes denoted as BS in this document), proposed by Fischer Black and Myron Scholes in 1973.

Building on earlier ideas of continuous-time price evolution from *Louis Bachelier* and rigorous stochastic calculus introduced by *Paul Samuelson*, *F. Black* and *M. Scholes* derived a closed-form solution for pricing European options under the assumptions of log-normal asset returns, constant volatility, and a risk-free interest rate.

Shortly thereafter, Robert Merton recast their result within a more general arbitrage-free framework and formalized the dynamic hedging argument, giving rise to what is often called the Black–Scholes–Merton model.



Figure 9: Fisher Black, Myron Scholes and Robert Merton

Although F. Black had passed away before the award, Scholes and Merton received the *1997 Nobel Prize in Economic Sciences* for their pioneering work.

The elegance and practicality of their formula not only transformed option markets but also laid the groundwork for decades of quantitative extensions, relaxing assumptions, incorporating stochastic volatility, jumps, and other real-world complexities.

³¹Bernard Lapeyre, Monte-Carlo methods and stochastic algorithms — <https://cermics.enpc.fr/~bl/monte-carlo/poly.pdf>

Note that the mathematical tools underpinning the Black–Scholes framework are highly sophisticated and draw upon foundational theories in stochastic processes, early mathematical finance, probability theory, stochastic calculus, martingale theory and continuous-time finance. Respectively, these areas were notably advanced by *Norbert Wiener*, *Louis Bachelier*, *Andrey Kolmogorov*, *Kiyosi Itô*, *James Laurie Snell* and *Robert Merton*, building on the work of many predecessors who have gone down in history.

Pre-Black-Scholes area, a heuristic approach -

Before the formalization of the Black-Scholes model, option pricing was largely guided by intuition, experience, and heuristic methods rather than rigorous mathematical frameworks.

Traders and analysts relied on empirical rules, historical averages, and ad-hoc adjustments to estimate the value of options.

The concept of arbitrage existed but was not formally embedded into pricing models. One early milestone was the work of *Louis Bachelier* in 1900, who introduced the use of *Brownian motion* to model stock prices and derived a formula for pricing options. However, his model assumed that prices followed a normal distribution, which led to the possibility of negative prices, an unrealistic feature that limited its adoption.

Throughout the mid-20th century, various practitioners developed pricing methods based on discrete-time models and replicating portfolios, but these remained fragmented and lacked the consistency and analytical power that the continuous-time stochastic calculus of Black-Scholes would later bring.

The pre-Black-Scholes era was thus characterized by approximations and practical ingenuity, setting the stage for the more systematic and theoretically grounded models that followed.

3.1 Price dynamics

The Black-Scholes model relies on a central assumption regarding the dynamics of the underlying asset price. It belongs to a category of model called "diffusion models", as prices evolve smoothly over time with random fluctuations.

It is assumed to follow a *geometric Brownian motion*, modeled by the following stochastic differential equation:

$$dS_t = \mu S_t dt + \sigma S_t dW_t \quad (1)$$

In the chosen notation, the asset price S_t is nothing more than the asset price at time t , and could also be written $S_t = S(t)$.

Note that μ represents the expected rate of return, σ is the volatility (assumed constant), and W_t is a standard Brownian motion i.e., a stochastic process with independent, stationary, normally distributed increments, with

$W_0 = 0$.

Here, the constant μ represents the *drift* or expected rate of return of the asset, while σ denotes the *volatility*, measuring the magnitude of random fluctuations driven by the Brownian motion W_t .

Now that we wrote this equation, let's discuss the rationale behind it, by considering a simplified version, removing the Brownian term.

$$dS_t = \mu S_t dt \quad (2)$$

Both (1) and (2) have an initial condition $S(0) = S_0 > 0$, meaning that the stock price at initial time t_0 is positive.

The dt notation represents an infinitesimal increment of time. The term $\mu S_t dt$ captures the deterministic part of the asset's price evolution. It reflects the expected return over a small time interval.

In the equation (2), simplified from (1), an analytical solution exists.

The equation is an 'ODE' (ordinary differential equation) where the right-hand side function $f(t, S_t) = \mu S_t$ is continuous and Lipschitz-continuous with respect to S_t , by the **Picard-Lindelöf**³², also called **Cauchy-Lipschitz** theorem there exists a unique solution for this initial value problem.

Now that we know the solution is unique, let's express it analytically.

Integrating (2) on both side yields $\int \frac{1}{S_t} dS_t = \int \mu dt$

Thus, $\ln |S_t| = \mu t + C$, with $C \in \mathbb{R}$

Let's solve for C , setting $t = 0$: $\ln |S_0| = C$

Consequently, $\ln |S_t| = \mu t + \ln S_0$

Exponentiating both sides, we get:

$$\boxed{S_t = S_0 e^{\mu t}}$$

³²**Picard-Lindelöf theorem:** Let $f : [a, b] \times \mathbb{R} \rightarrow \mathbb{R}$ be continuous in both arguments and Lipschitz continuous in the second argument; i.e., there exists $L > 0$ such that for all $t \in [a, b]$ and all $x, y \in \mathbb{R}$,

$$|f(t, x) - f(t, y)| \leq L|x - y|.$$

Then, for any initial condition $S(t_0) = S_0$, there exists a unique solution $S(t)$ defined on some interval containing t_0 to the initial value problem

$$\frac{dS}{dt} = f(t, S), \quad S(t_0) = S_0.$$

The solution represents the price of an asset evolving over time t with a constant instantaneous rate of return μ . The parameter μ can be interpreted as the continuous growth rate of the asset's value, meaning that the asset price grows exponentially at this rate. Financially, this implies that the continuously compounded return over the time interval $[0, t]$ is μt , so the price at time t reflects the initial price multiplied by the exponential growth factor $e^{\mu t}$. This deterministic model captures the idea of steady, predictable growth (or decay, if $\mu < 0$) of the asset price in the absence of randomness or volatility.

Do gain a complete intuition on the equation (1) :

$$dS_t = \mu S_t dt + \sigma S_t dW_t \quad (1)$$

We know need to understand what this mysterious term is.

The Brownian motion was first observed in 1827 by the surgeon and botanist **Robert Brown**, who noticed that pollen particles suspended in water exhibited a seemingly erratic and perpetual motion under a microscope.



Figure 10: Robert Brown

This phenomenon was later understood to result from collisions with molecules of the surrounding fluid.

In mathematics, the *Brownian motion* (also called *Wiener process*, named after Norbert Wiener who formalized its mathematical construction) is modeled as a continuous-time stochastic³³ process $(W_t)_{t \geq 0}$, meaning a continuous sequence of random variables, where for each t , W_t is a random variable of a given law.

It is characterized by the following properties :

³³The term *stochastic* refers to systems or processes that involve randomness or probabilistic behavior. In contrast to deterministic systems, where future outcomes are fully determined by initial conditions, stochastic models incorporate random variables or processes such as Brownian motion

- $W_0 = 0$ almost surely;
- $W_t \sim N(0, t)$;
- The process has independent increments;
- The increment $W_{t+s} - W_t$ is normally distributed with mean 0 and variance s for any $s, t \geq 0$;
- The paths of W_t are almost surely continuous in t .

Sampling a path followed by such process yields graphs looking like :

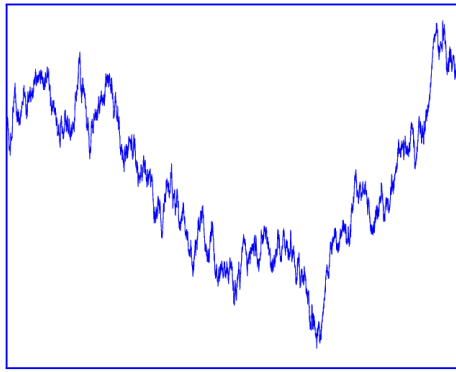


Figure 11: *1-dimensional* Brownian motion trajectory, drawn from a sample of normal random variables

Does that shape ring a bell?

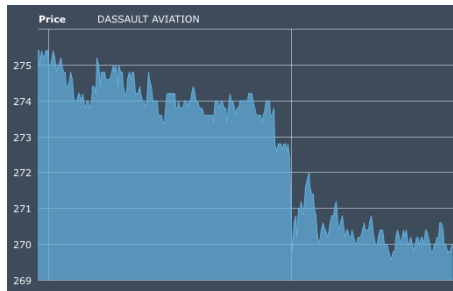


Figure 12: Dassault Aviation stock price - morning of the 1st of August 2025

Yes, financial asset prices look like Brownian trajectories. As the molecules in glass of water hitting unceasingly the grain of pepper, wiggling it at the surface in random directions, the financial asset prices go up and down tirelessly, hit by the constantly changing supply and demand, market and political events, personal sentiments.

Thus, the Brownian motion serves as the fundamental source of randomness in many models of financial markets, particularly in the modeling of asset prices.

Now that we defined the Brownian a little bit more, let's demystify the **volatility term** of the Black-Scholes dynamic.

$$dS_t = \mu S_t dt + \sigma S_t dW_t \quad (1)$$

The term dW_t denotes the infinitesimal increment³⁴ of a standard Brownian motion W_t .

It models the random fluctuations of the asset price and satisfies $dW_t \sim N(0, dt)$, meaning that it is normally distributed with zero mean and variance dt .

The volatility term $\sigma S_t dW_t$ thus represents the stochastic component of the price dynamics. The coefficient σ is the volatility, which scales the magnitude of the randomness, and the fact that it is multiplied by S_t implies that the randomness is proportional to the current asset price.

The σ parameter can be seen as a measure of the market's perceived uncertainty about the future evolution of the asset price. A higher volatility indicates greater uncertainty and larger potential fluctuations in the price path, while a lower volatility suggests a more stable asset. In the Black-Scholes framework, this volatility is assumed to be constant over time, which greatly simplifies the mathematical treatment but diverges from observed market behavior.

The addition of deterministic drift and stochastic noise forms the *geometric Brownian motion*.

Using Itô's lemma³⁵, one can compute explicitly the law followed by the asset price at time T . You will see how handy this is later on.

$$S_T = S_0 e^{(\mu - \frac{\sigma^2}{2})T + \sigma W_T} \quad (3)$$

Since $W_T \sim N(0, T)$, it follows that $\log S_T$ is normally distributed. Thus, S_T follows a log-normal distribution, with

$$\log S_T \sim N\left(\log S_0 + \left(\mu - \frac{\sigma^2}{2}\right)T, \sigma^2 T\right).$$

³⁴The notation dW_t should be understood in the framework of stochastic calculus, specifically Itô calculus.

Although it resembles a derivative, it is not one in the classical sense because the Brownian motion W_t is almost surely nowhere differentiable. In fact, standard analysis does not allow us to define $\frac{dW_t}{dt}$.

However, stochastic calculus provides a rigorous meaning to expressions involving dW_t , by interpreting them as limits of stochastic processes. For instance, dW_t can be thought of as a random variable with distribution $N(0, dt)$, and integrals of the form $\int_0^t f(s) dW_s$ are well-defined through the Itô integral, which relies on adapted processes and square-integrability conditions rather than classical notions of differentiability.

Since this is not a stochastic calculus course, find more details in : Brownian Motion, Martingales, and Stochastic Calculus, Le Gall - Springer 2016 or for French speakers: Introduction au calcul stochastique appliqué à la finance, Lamberton & Lapeyre - 2013

³⁵The key idea is to apply Itô's lemma to the logarithm of the asset price. This transformation linearizes the equation, turning the multiplicative noise in S_t into additive noise in $\log S_t$. The resulting equation describes a process with a deterministic drift and a Brownian motion term. Since Brownian motion is normally distributed, the solution for $\log S_T$ is a normal random variable. Exponentiating gives the distribution of S_T as log-normal.

This fully characterizes the law of S_T . Find the proof of this result in the Appendix.

Quiz (4) - Black-Scholes modelling

Each question has one or many possible answers.

1. A standard Brownian motion W_t has the following properties:
 - ☐ Continuous paths
 - ☐ Stationary increments
 - ☐ Independent increments
2. In the BS model, the stock price dynamics are given by:
 - ☐ $dS_t = \mu S_t dt + \sigma dW_t$
 - ☐ $dS_t = \mu S_t dt + \sigma S_t dW_t$
 - ☐ $dS_t = \mu dt + \sigma S_t dW_t$
3. Under the risk-neutral measure $\mathbb{Q}$, the drift of the stock price becomes:
 - ☐ Zero
 - ☐ The risk-free rate r
 - ☐ The historical mean return μ
4. The solution to the BS SDE for S_t is:
 - ☐ $S_t = S_0 e^{rt}$
 - ☐ $S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$
 - ☐ $S_t = S_0 + \mu t + \sigma W_t$
5. The variance of Brownian motion W_t is:
 - ☐ $\sqrt{t}$
 - ☐ t
 - ☐ t^2
6. In the BS model, what is the distribution of the log-returns of the underlying asset over a time interval?
 - ☐ Uniform distribution
 - ☐ Normal distribution
 - ☐ Exponential distribution
7. In the BS model, volatility is assumed to be:
 - ☐ Jump-driven
 - ☐ Deterministic and constant
 - ☐ Stochastic and mean-reverting
8. Which process is equal in law to $(B_t)_t$?
 - ☐ $(\sigma B_t)_t$
 - ☐ $(tB_{\frac{1}{t}})_t$
 - ☐ $(B_{t^2})_t$
9. What is the most important problem with Bachelier's model for asset prices, where the price follows $S_t = S_0 + \sigma W_t$?
 - ☐ It ignores interest rates
 - ☐ It allows negative asset prices
 - ☐ It assumes constant relative volatility

10. Volatility of an asset can be described as:
- ☐ How nervously traders check their screens at 3:00 pm
 - ☐ The market's expectation of how much the asset price will fluctuate
 - ☐ The square root of the expected value of the squared deviation of a random variable from its mean

3.2 Vanilla option analytical formula

As discussed in previous sections, a vanilla option refers to a standard European call or put option, which gives the holder the right, but not the obligation, to buy or sell an underlying asset at a predetermined strike price on a fixed maturity date.

Let's just recall the payoff of a call option at maturity T :

$$(S_T - K)_+ = \begin{cases} S_T - K & \text{if } S_T \geq K \\ 0 & \text{otherwise} \end{cases}$$

The payoff of the put option is $(K - S_T)_+$

Let's dive into the biggest perk of the Black-Scholes framework that revolutionized the world of finance: expressing the price of a European option using a closed-form formula to compute premium effortlessly.

Under a certain probability measure³⁶ denoted $\mathbb{Q}$, the price of a European call option at time $t = 0$ with maturity T and strike K is given by the discounted expectation of its payoff:

$$C_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}} [(S_T - K)_+].$$

The exponential factor e^{-rT} is the *discount factor*, which accounts for the time value of money: a payoff received at time T must be discounted back to time 0 using the continuously compounded risk-free rate r . Without this term, the expectation would simply give the average payoff at maturity, but multiplying by e^{-rT} adjusts it to obtain the present value of the option. Intuitively, one euro available today can be invested risk-free to yield more than one euro tomorrow, which is why future cash flows are worth less than immediate ones.

Using (3), and working under $\mathbb{Q}$, we have:

$$S_T = S_0 e^{\left(r - \frac{\sigma^2}{2}\right)T + \sigma W_T^{\mathbb{Q}}}, \quad \text{with } W_T^{\mathbb{Q}} \sim N(0, T).$$

We can write the payoff as a function of a normal variable:

$$(S_T - K)_+ = \left(S_0 e^{\left(r - \frac{\sigma^2}{2}\right)T + \sigma \sqrt{T}Z} - K \right)^+, \quad \text{where } Z \sim N(0, 1).$$

³⁶The measure $\mathbb{Q}$, called the risk-neutral measure, is an artificial but extremely useful probability measure under which we assume that all assets grow on average at the risk-free interest rate r . This simplifies pricing because, instead of trying to model investors' actual expectations and risk preferences, we can calculate the expected payoff of an option as if investors were risk-neutral—hence the name. This change of perspective allows us to compute fair option prices while still reflecting the actual market dynamics.

Mathematically, the measure $\mathbb{Q}$ is obtained from the real-world probability measure $\mathbb{P}$ through a change of measure (via Girsanov's theorem). The change of measure adjusts the drift of the stochastic differential equation followed by the asset price: under $\mathbb{P}$, the drift is the actual expected return μ , but under $\mathbb{Q}$, it becomes the risk-free rate r . This guarantees that the discounted asset price process $e^{-rt}S_t$ is a martingale, which is the key condition ensuring absence of arbitrage

With the notation $(x)^+ = \max(x, 0)$ we denote the positive part of x , which is equal to x if x is positive and 0 otherwise.

Define:

$$X := \left(S_0 e^{\left(r - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}Z} - K \right)^+,$$

where $Z \sim N(0, 1)$

By definition of the expectation of a function $g(Z)$ of a continuous random variable Z with density $\phi(z)$, we have

$$\mathbb{E}[g(Z)] = \int_{-\infty}^{+\infty} g(z) \phi(z) dz.$$

Applying this to the function

$$g(z) := \left(S_0 e^{\left(r - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}z} - K \right)^+,$$

the expected value becomes

$$\mathbb{E}^{\mathbb{Q}}[X] = \int_{-\infty}^{+\infty} \left(S_0 e^{\left(r - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}z} - K \right)^+ \phi(z) dz.$$

where $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ is the standard normal density. Since the integrand is zero whenever the expression inside the parentheses is negative, we can restrict the integral to the domain where:

$$S_0 e^{\left(r - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}z} > K.$$

Solving for z gives the threshold:

$$z > \frac{\log(K/S_0) - \left(r - \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} =: z^*.$$

So we rewrite the expectation as:

$$\mathbb{E}^{\mathbb{Q}}[X] = \int_{z^*}^{+\infty} \left(S_0 e^{\left(r - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}z} - K \right) \phi(z) dz.$$

We now split the integral into two parts:

$$\mathbb{E}^{\mathbb{Q}}[X] = S_0 \int_{z^*}^{+\infty} e^{\left(r - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}z} \phi(z) dz - K \int_{z^*}^{+\infty} \phi(z) dz.$$

The second integral is straightforward:

$$\int_{z^*}^{+\infty} \phi(z) dz = N(-z^*) = N(d_2),$$

since $z^* = -d_2$.

For the first term, we compute:

$$\int_{z^*}^{+\infty} e^{\left(r - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}z} \phi(z) dz = e^{\left(r - \frac{\sigma^2}{2}\right)T} \int_{z^*}^{+\infty} e^{\sigma\sqrt{T}z} \phi(z) dz.$$

Let's simplify the integrand:

$$e^{\sigma\sqrt{T}z} \phi(z) = \frac{1}{\sqrt{2\pi}} e^{\sigma\sqrt{T}z - \frac{z^2}{2}}.$$

Complete the square in the exponent:

$$\sigma\sqrt{T}z - \frac{z^2}{2} = -\frac{1}{2} \left(z^2 - 2\sigma\sqrt{T}z \right) = -\frac{1}{2} \left(z - \sigma\sqrt{T} \right)^2 + \frac{\sigma^2 T}{2}.$$

So we get:

$$e^{\sigma\sqrt{T}z} \phi(z) = \frac{1}{\sqrt{2\pi}} e^{\frac{\sigma^2 T}{2}} e^{-\frac{1}{2}(z - \sigma\sqrt{T})^2}.$$

Now substitute into the integral:

$$\int_{z^*}^{+\infty} e^{\sigma\sqrt{T}z} \phi(z) dz = e^{\frac{\sigma^2 T}{2}} \int_{z^*}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z - \sigma\sqrt{T})^2} dz.$$

This is the integral of a normal density centered at $\sigma\sqrt{T}$, so we perform a change of variable:

$$u = z - \sigma\sqrt{T}, \quad du = dz.$$

The lower limit becomes $z^* - \sigma\sqrt{T} = -d_1$. So we get:

$$\int_{z^*}^{+\infty} e^{\sigma\sqrt{T}z} \phi(z) dz = e^{\frac{\sigma^2 T}{2}} \int_{-d_1}^{+\infty} \phi(u) du = e^{\frac{\sigma^2 T}{2}} N(d_1).$$

Putting everything together:

$$\mathbb{E}^{\mathbb{Q}}[(S_T - K)^+] = S_0 e^{\left(r - \frac{\sigma^2}{2}\right)T} e^{\frac{\sigma^2 T}{2}} N(d_1) - KN(d_2) = S_0 e^{rT} N(d_1) - KN(d_2).$$

Finally, multiplying by the discount factor e^{-rT} , we obtain:

$$C_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+] = S_0 N(d_1) - Ke^{-rT} N(d_2)$$

which is the Black-Scholes formula for a European call option.

The derivatives of this formula, proving the sensitivities of the option price to variations in market price, volatility, rates, etc... will be discussed in the [Greeks](#) section, Risk Management chapter.

Put-call parity is a fundamental relationship between the prices of European call and put options with the same strike K and maturity T , and is given by:

$$C_0 - P_0 = S_0 - Ke^{-rT},$$

where C_0 and P_0 are respectively the prices of the European call and put options at time $t = 0$.

This relation comes from a no-arbitrage argument: holding a call and selling a put with the same strike and maturity is equivalent to entering a forward contract to buy the asset at price K at time T . This forward contract is worth $S_0 - Ke^{-rT}$ under the risk-neutral measure.

Arbitrage refers to the possibility of making a profit without taking any risk or investing any capital. More precisely, an arbitrage opportunity exists if a trading strategy requires zero initial investment, has zero probability of incurring a loss, and has a strictly positive probability of making a profit. In an efficient and competitive market, such opportunities are assumed to be exploited immediately, driving prices back to a level where arbitrages do not exist anymore.

From the put-call parity, we isolate P_0 to get the put price:

$$P_0 = C_0 - S_0 + Ke^{-rT}.$$

Now using the Black-Scholes formula for the call price:

$$C_0 = S_0N(d_1) - Ke^{-rT}N(d_2),$$

and substituting into the put-call parity, we obtain:

$$P_0 = (S_0N(d_1) - Ke^{-rT}N(d_2)) - S_0 + Ke^{-rT}.$$

Rearranging the terms:

$$P_0 = -S_0(1 - N(d_1)) + Ke^{-rT}(1 - N(d_2)).$$

Recalling that $1 - N(x) = N(-x)$, we get the Black-Scholes formula for a European put:

$$P_0 = Ke^{-rT}N(-d_2) - S_0N(-d_1)$$

which is the price of a European put option under the Black-Scholes model.

The chapter "Risk Management" will cover the *Greeks*, meaning the partial derivatives of the option price with respect to various model parameters. These sensitivities quantify how the value of an option changes in response to small changes in underlying factors such as the price of the asset, volatility, time to maturity, etc...

Quiz (5) - Vanilla Option Pricing Theory

Each question has one or many possible answers.

1. In the derivation of the Black-Scholes formula, after the change of variables, the PDE reduces to which standard form?
 - ☐ Wave equation
 - ☐ Heat equation
 - ☐ Poisson equation
2. The risk-neutral valuation formula for a European call option is:
 - ☐ $\mathbb{E}[(S_T - K)^+]$ under real-world probability
 - ☐ $\max(S - K, 0)$ at time 0
 - ☐ $e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+]$
3. In the Black-Scholes call formula $C = S_0 \Phi(d_1) - K e^{-rT} \Phi(d_2)$, d_1 and d_2 come from:
 - ☐ Approximating the option by linear functions
 - ☐ Discretizing the PDE
 - ☐ Transforming the lognormal integral into the standard normal CDF
4. How to "easily" compute the price of the European put option?
 - ☐ The put price is the inverse of the call price S
 - ☐ Using the call-put parity
 - ☐ From the hypothesis of nonexistence of arbitrages $S = \infty$
5. The factor e^{-rT} in the analytical formula represents:
 - ☐ Discounting expected payoff at the risk-free rate
 - ☐ Transforming the PDE into a heat equation
 - ☐ Adjusting for volatility
6. To compute the BS option prices numerically, one can evaluate:
 - ☐ It is impossible to evaluate numerically
 - ☐ The integral over S_T analytically only
 - ☐ The standard normal CDF $\Phi(d)$ using numerical approximation
7. When solving the option price, yielding the Black-Scholes formula, the evaluated integral can be interpreted as:
 - ☐ Present value of the strike price
 - ☐ Variance of the payoff
 - ☐ Expected value of S_T conditioned by $S_T > K$
8. In the BS framework, a European call option gives the holder the right to:
 - ☐ Receive dividends of the underlying before maturity
 - ☐ Buy the underlying asset at strike T at maturity K
 - ☐ Buy the underlying asset at strike K at maturity T
9. The payoff of a vanilla call option at maturity is:
 - ☐ $(S_T - K)_+$
 - ☐ $S_T - K$
 - ☐ $\max(S_T - K, 0)$

10. The essential theorem allowing to obtain the measure $\mathbb{Q}$ is named after:
- ☐ Igor Girsanov
 - ☐ Andreï Kolmogorov
 - ☐ Vladimir Molotov

3.3 Limitations

The Black-Scholes model has been revolutionary in providing a closed-form analytical solution for European option pricing and hedging under a set of elegant assumptions. Even though this model is incredibly powerful, it has notable flaws. Let's discuss them in this section.

One particularly important use of the Black-Scholes formula is the following: given an observed market price of an option, one can invert the formula to find the unique volatility input σ that would make the model price match the market price.

This leads to the concept of *implied volatility*, defined as the solution σ_{impl} to the equation: (Call option example)

$$C_{\text{market}} = C^{\text{BS}}(S_0, K, T, r, \sigma_{\text{impl}}),$$

where C_{market} is the observed call market price³⁷ of the option, and C^{BS} is the function that prices Black-Scholes calls given all the mandatory parameters.

However, one of the model's main limitations lies in the assumption of *constant volatility*. In the model, the volatility parameter σ is treated as a fixed input, independent of both time and the level of the underlying asset price. This assumption is at odds with observed market behavior. In reality, implied volatilities vary across strikes and maturities, forming structures known as the *volatility smile* and *skew*.

This empirical inconsistency indicates that market participants anticipate different levels of uncertainty (volatility) depending on the option's moneyness (strike value) and time to maturity – features the original Black-Scholes model fails to capture.

The concepts of smile, skew and surface are capital to understand. Let's dive in them a little more, by analyzing some option market data.

³⁷In practice, options are quoted in terms of implied volatility rather than absolute monetary prices. Market makers provide bid and ask quotes directly in implied volatilities, because implied volatility isolates the market's view on future uncertainty and is less sensitive to small changes in the spot price. Quoting in dollars would require continuous adjustments for even minor spot movements, whereas implied volatility remains relatively stable, since it reflects the market-implied variance of the asset's distribution at maturity, which does not fluctuate instantaneously with short-term spot noise

Euro Stoxx 50 Pr Index OMON Related Functions Menu									
SX5E 5201.06 +35.46 -- / --									
At 09:29 d 0 5170.48 H 5201.87 L 5170.48 Prev 5165.60									
SX5E Index 95 Actions 90 Export 97 Settings									
Euro Stoxx 50 Pr 15201.06 35.46 .6864% / Hi 5201.87									
Center 5200.01 Strikes 18 Exp 15-Aug-25 Exch Eurex Germany 90 Events Calendar EVIS »									
Calc Mode									
Center Strike Calls/Puts Calls Puts Term Structure Moneyness									
Bid Mid Ask									
Tenor									
		80	85	90	95	97.5	100		
1W		49.44y	40.92y		25.68	24.19	18.30		
2W		48.75	39.42	31.27	23.70	20.34	17.33		
3W		45.53	37.15	29.80	22.96	19.85	17.11		
1M		40.85	32.39	26.19	21.36	18.83	16.58		
2M		32.14	27.47	23.40	19.68	17.80	16.11		
3M		29.52	25.75	22.20	18.94	17.43	16.09		
6M		25.62	23.01	20.34	18.12	17.03	16.11		
9M		23.43	21.32	19.27	17.52	16.68	15.91		
1Y		22.57	20.64	18.90	17.45	16.70	16.04		
18M		21.52	19.97	18.55	17.49	16.85	16.40		
2Y		20.93y	19.62y	18.45y	17.39y	16.87y	16.39y		

Figure 13: Bloomberg SX5E call option quotations for various strikes and maturities

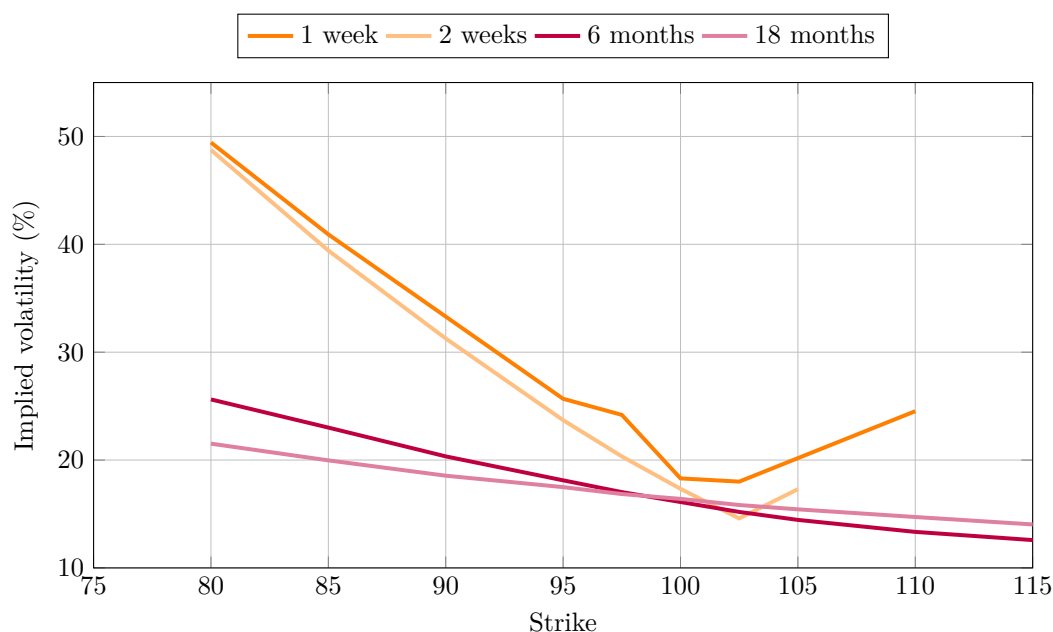


Figure 14: Volatility smile & skew on SX5E call option quotations

As shown on this graph, the implied volatility is a **non-constant** function of the option maturity and strike. The phenomenon of the smile is characterized by higher implied volatility for deep in-the-money and out-of-the-money³⁸ options,

³⁸An option is said to be *in the money* if exercising it would be profitable: for a call option, this means the strike is below the current spot price, while for a put option, the strike is above the spot. Conversely, an option is *out of the money* if exercising it would not be profitable — that is, a call with a strike above the spot or a put with a strike below the spot. An option is *at the money* when the strike is approximately equal to the current spot price

reflecting market expectations of large price moves.

This is a strong evidence that the Black-Scholes constant volatility assumption is not verified in practice.

In our example, we also observe a **skew** for short term maturities. The skew refers to the asymmetry in implied volatility across strikes, usually reflecting higher implied volatility for low strikes (put side) compared to high strikes (call side).

In other words, market participants tend to buy protection against downside moves more aggressively than against upside.

For more details about the volatility surface, find this excellent reference by P. Tankov.³⁹

³⁹https://masterfinance.math.univ-paris-diderot.fr/attachments/article/39/poly_volatilite.pdf

4 Pricing in Practice

This section will be treated in the form of a Python workshop, using the following Jupyter Notebook. You'll find the correction on the repository too.

We will begin with a Brownian Motion Simulation, introducing stochastic processes through the simulation of Wiener paths.

Next, we will perform a terminal price distribution analysis by simulating the terminal asset price S_T under Geometric Brownian Motion and comparing the empirical distribution with the theoretical lognormal distribution predicted by the Black–Scholes model.

We will then move to Black–Scholes analytical pricing, implementing the closed-form solution for European options.

After that, we will simulate full asset price trajectories under the Black–Scholes dynamics.

Finally, we will address exotic option pricing via Monte-Carlo, pricing path-dependent options such as asian and barrier options.

We will also study Monte Carlo convergence.

<https://github.com/SK8gh/EFREI/blob/main/Training/Python%20Core.ipynb>

5 Greeks

Risk management is a fundamental aspect of financial decision-making that involves identifying, measuring, and controlling potential losses arising from participant's positions confronted to market fluctuations.

In trading and investment, uncertainties related to price movements, volatility, interest rates, and other factors can significantly impact portfolio value. Effective risk management seeks to understand these risks, quantify their potential effects, and implement strategies to mitigate adverse outcomes. This section provides an overview of the key concepts and tools used to manage these financial risks, laying the groundwork for more detailed discussions on sensitivities and hedging techniques.

Hedging consists in executing a strategy to reduce or eliminate the potential negative impact of adverse movements in a portfolio. It involves taking an offsetting position, often in a related security or derivative, that gains value when the original position loses value.

The goal of hedging is not theoretically to make a profit but to protect against losses and stabilize financial outcomes under uncertain market conditions.

As seen previously, the price of an option in the Black-Scholes framework depends on several variables: the price of the underlying asset, time to maturity, volatility, interest rate.

The greeks measure how sensitive the option price is to small changes in these inputs.

5.1 Derivatives of the Black-Scholes formula

A *position* refers to the variety and amount of financial instruments, that a trader currently holds. Positions can be *long* or *short*. Traders rely heavily on the *Greeks* to understand and control the risks embedded in their positions.

Rather than focusing solely on the nominal value of an option or portfolio, they monitor how the value is expected to evolve as market conditions change. This allows them to hedge specific types of risk (e.g. price movements, volatility changes, time decay) more precisely.

Each Greek corresponds to a partial derivative of the option price with respect to one of the input parameters. In practice, they are used to:

- Construct hedging strategies that isolate or neutralize specific risks.
- Estimate the impact of small market movements on a portfolio's value.
- Perform scenario analysis and stress testing.

The most commonly used Greeks are presented below.

Delta -

Delta measures the sensitivity of the option price to changes in the price of the underlying asset. It is defined as:

$$\Delta = \frac{\partial V}{\partial S}$$

where V is the value of the option and S is the spot price of the underlying.

For European options in the Black-Scholes model, the delta is given by:

$$\begin{cases} \Delta_{\text{call}} = N(d_1) \\ \Delta_{\text{put}} = \Delta_{\text{call}} - 1 \end{cases}$$

where $N(\cdot)$ is the cumulative distribution function of the standard normal distribution, and d_1 is defined in the [Black-Scholes section](#).

We often interpret the delta as the probability that the option will finish in the money, meaning that it will be worth exercising it.

Indeed, the risk-neutral probability of the option expiring in the money is $\mathbb{Q}(S_T > K) = N(d_2)$.

For a call, while $\Delta_{\text{call}} = N(d_1) \neq N(d_2)$ in general, the two values become close when the time to maturity is small or the interest rate is low.

In such cases, we have:

$$\Delta_{\text{call}} = N(d_1) \approx N(d_2) = \mathbb{Q}(S_T > K)$$

This approximation justifies the common intuition that the delta reflects the likelihood that the option will be exercised.

Parameters: $S = 100$, $K = 100$, $\sigma = 0.2$, $r = 0.01$, $T = 0.01$

$$\begin{cases} d_1 \approx 10^{-2} \\ d_2 = 0 \end{cases} \quad \begin{cases} \Delta_{\text{call}} = N(d_1) \approx 0.50398 \\ \mathbb{P}(S_T > K) = N(d_2) = 0.5 \end{cases}$$

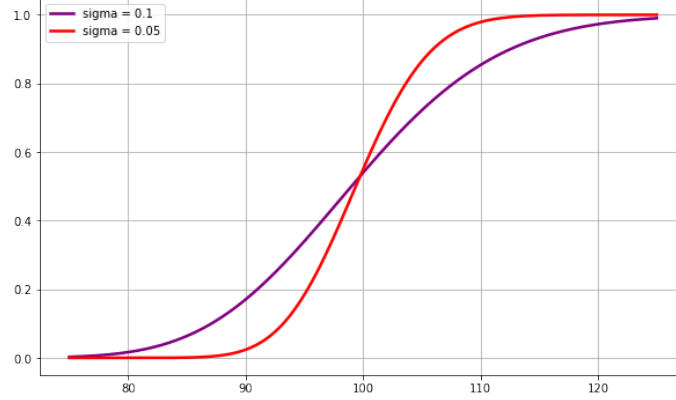


Figure 15: Call delta exposure as a function of the strike

As shown on the Fig. above, the delta is steeper when the volatility decreases. Indeed, under a very low volatility condition, if the spot goes up or down, it is more likely to stay where it is. Thus, the option price is sensitive to small moves. When the volatility decreases, the payoff of a European option looks more and more like a digital option payoff. (Refer to the Appendix for more details about digital options)

$$\text{Payoff}_{\text{digital}}(S_T) = \mathbf{1}_{\{S_T > K\}} = \begin{cases} 1 & \text{if } S_T > K \\ 0 & \text{if } S_T \leq K \end{cases}$$

Payoff of a digital call option. The option pays a fixed amount (here normalized to 1) if the underlying asset price at maturity S_T exceeds the strike K , and 0 otherwise.

Delta is widely used to build *delta-neutral* portfolios, where the overall sensitivity to small spot price movements is minimized.

Let's provide a concrete example of **delta-hedging**.

Consider a trader who is short (in other words, he sold the call) of a European call option of nominal \$1M, on the underlying stock *Boeing*, currently trading at 223\$ (August 2025), meaning that the option covers \$1M worth of stocks, $\frac{10^6}{223} = 4484$ stocks at current spot price.

The analytic delta formula lets the trader know his option's $\Delta_{\text{call}} = 0.60$, and since he sold the call, we will adjust his delta by a -1 factor. $\Delta_{\text{call}} = -0.60$ making him *short delta* because his $\Delta_{\text{total}} = \Delta_{\text{call}} < 0$. Otherwise, we would

say he is *long delta*. His call position will appreciate if the Boeing stock loses value.

To hedge this position and make the position delta-neutral, the trader needs to **buy some delta**, which he can do by buying $4484 \cdot \Delta_{call} = 4484 \times 0.6 \approx 2690.4$ shares of the underlying stock (since the stock has a delta of 1 per share)

Let's compute the trader's P&L (\$) for a $\pm 1\%$ stock move.

Spot move	Model call P&L	Stock P&L	Total P&L
-1%	+6,000	-5,999.592	-0.408
+1%	-6,000	+5,999.592	+0.408

Table 6: P&L (\$) of the delta-hedged position for a 1% spot move

The *Table 6* shows that the total P&L ≈ 0 no matter where the stock goes, the position is **theoretically delta-hedged**.

Note, however, that as the spot price, volatility, or time to maturity change, the option's delta will also change. Therefore, if a trader needs to keep his position delta hedged, the hedge must be adjusted — this practice is known as *dynamic delta hedging*.

Also note that the trader used the BS model to evaluate $\Delta_{call} = -0.6$ but in practice, the "realized" delta could be different. Indeed, since the model does not picture the reality perfectly, the greeks computed from the Black-Scholes closed formulas are always inaccurate.

More generally, everything any model says is inaccurate. The actual prices of assets priced using models are in reality at least slightly different than the intrinsic value of the products. This is totally normal, to give a more tangible analogy: no model of weather prediction can predict the weather perfectly, they can estimate it quite correctly though, which can be perceived as "enough".

Gamma -

Gamma measures the rate of change of delta with respect to changes in the price of the underlying asset. It is the second derivative of the option value with respect to the spot price:

$$\Gamma = \frac{\partial^2 V}{\partial S^2} = \frac{\partial \Delta}{\partial S}$$

Gamma tells the trader how much the delta of an option will change if the underlying moves.

In the Black-Scholes model, the gamma of a European call or put option is the same, and given by:

$$\Gamma_{call} = \Gamma_{put} = \frac{N'(d_1)}{S\sigma\sqrt{T}} \quad (4)$$

Where S is the current spot price, and T the time to maturity.

Gamma is *always positive* when buying vanilla options, and highest for at-the-money options with short maturities.

Mathematically, the positivity of the gamma it is caused by the convexity of the options payoff. A financial intuition would be that when the spot moves in a favorable direction, the probability of exercise goes up (hence the delta, as seen before it is often said to approximate the likelihood to end-up in the money)

Intuitively, high gamma means that delta will change rapidly with spot moves, so the trader must re-hedge more frequently to maintain a delta-neutral position. On the other hand, low gamma implies delta is relatively stable.

Gamma is crucial for risk management: even if the portfolio is perfectly delta-neutral now, if it has a large gamma exposure, it might become significantly non-neutral after a small move in the underlying. This gives rise to **gamma risk**.



Figure 16: Average option trader sayings

Options market makers often try to manage both delta and gamma exposures. While delta can be hedged using the underlying asset, gamma cannot. The only way to hedge gamma is to enter new option contracts.

In practice, option market makers rarely gamma-hedge perfectly their position for the reason we just evoked. Usually, their activity yield a certain greek exposure, that they mitigate, tweaking their greeks to follow strategies or sentiments.

Using (4) let's choose some model, spot and maturity parameters to approximate d_1 and gain some intuition on the sign of the gamma for call & put options.

With $S = 100$, $\mu = 0.05$, $\sigma = 0.3$, $T = 1$ we can plot the gamma of a call of strike K , under the BS model.

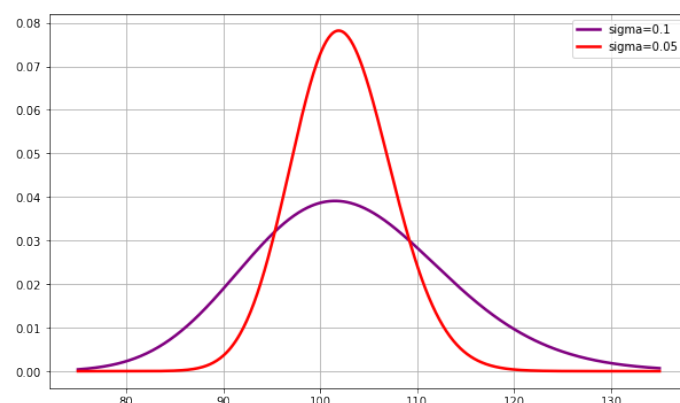


Figure 17: Call gamma exposure as a function of the strike

The gamma of an option peaks near the spot price of the underlying asset. To gain intuition about the shape of the gamma curve, it is useful to recall that the delta of a call option is often interpreted as the risk-neutral probability that the option finishes in the money, meaning that it will be exercised.

For an at-the-money call option, this probability is close to 0.5, and even small movements in the spot price cause large changes in delta — because the likelihood of ending in the money changes rapidly. This high sensitivity of delta to the spot price around the strike is precisely what gamma measures.

As a result, gamma is highest when the option is ATM, and decreases as the option becomes deep in-the-money or far out-of-the-money, where delta tends to plateau near 1 or 0, respectively.

Furthermore, as shown by Fig. 7, the lower the volatility the higher gamma peaks, exactly for the same reason the delta steepens around the money. It also get more peaky when the maturity approaches. *Try figuring out why?*

Gamma is often called the "acceleration", in the sense that it measures how quickly the delta of an option changes as the spot varies. Just as acceleration is the rate of change of velocity in physics, gamma is the rate of change of delta with respect to the spot price.

5.2 Delta & gamma management in practice -

Let's *put* ourselves once again in the shoes of Audrey, an options trader, who holds a very simple delta-hedged position having a single call option, either long or short.

Direction	Delta Hedge	Delta	Gamma
Bought the call	Short Underlying	Positive	Positive
Sold the call	Long Underlying	Negative	Negative

Imagine Audrey works on a market-making desk with a mandate⁴⁰ requiring her to remain delta-neutral throughout the life of the contract. This means she must regularly adjust her hedge whenever the underlying price moves. These hedge adjustments can be thought of as *automatic take-profit or stop-loss orders*: they are triggered not by directional speculation, but by the need to (1) comply with delta exposure limits, and (2) lock in profits or limit losses when the spot moves significantly.

Let's explore how this plays out in practice.

Example: Short call with negative gamma Audrey sells a Boeing call option with a notional value of \$1 million. Initially, she delta-hedges by buying 2,690 Boeing shares at \$223 each (see [delta-hedging example](#) for details). At this point, her total delta is roughly equal to 0.

Now suppose the spot rises to \$224. Because she is short the call, she has negative gamma: as the spot rises, her delta decreases, so her hedge becomes too long. She must therefore *buy* some of her stock, now at \$224 instead of the \$223 she was initially paying to delta-hedge.

Later, if the spot falls back to \$223, her delta increases (remember, she is gamma negative, thus when the spot moves in a direction, the delta moves in the other direction) She therefore *sells* shares at \$223 to restore neutrality. Here, she has bought high and sold low, generating a negative P&L purely from gamma and delta management.

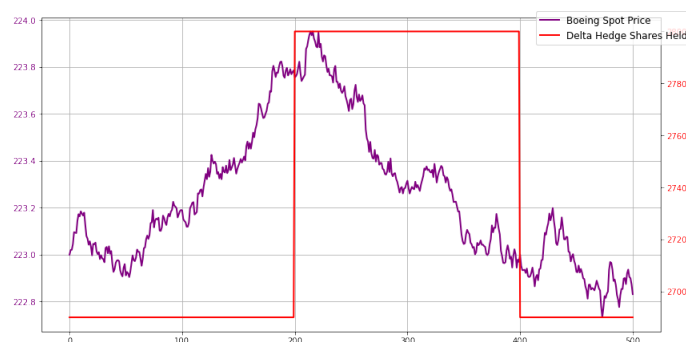


Figure 18: Audrey's delta hedge position & Boeing stock price

Note that in the graph above, the quantity bought to adjust the delta hedge must be evaluated by computing how much the delta decreased after the spot increase.

If instead the spot drops first and then goes up again, the mechanism is the same but in reverse: she sells stocks on the dip (to stay delta-hedge) and buys them back when the price recovers, again losing money from volatility.

In the example above, Audrey's timing was loopy, and each delta-hedging adjustment coincided exactly with local price highs or lows, as shown on the

⁴⁰ A trading mandate issued by employers typically includes limits on risk exposure, required hedging strategies, and overall trading objectives

Fig. 8. In reality, deciding when to adjust, and thus when to “take profit” or “stop loss” on the hedge, is much harder, and execution costs matter. Utilizing this “strategy”, traders can make or loss money depending on their timing and obligations of delta adjustments.

Example: Long call with positive gamma. If Audrey had instead *bought* the call, her hedge would be the opposite: she would initially *sell* stock to stay delta-neutral, and her gamma would be positive.

When the spot rises, her delta increases, she sells the stock at a higher price to restore neutrality. When the spot falls back, she buys back at a lower price – a profit-taking move.

Summary table – Impact of market moves on delta-hedging P&L

Gamma Exposure	Volatility	Dynamic Delta-Hedging	P&L Outcome
Positive	High	Buy low, sell high	Profit
Negative	Low	Few adjustments	Near zero
Negative	High	Buy high, sell low	Loss
Positive	Low	Few adjustments	Near zero

Table 7: P&L impact from delta-hedging adjustments depending on gamma exposure and market volatility. To remember: when hedging delta dynamically, you profit from volatility when $\Gamma > 0$

Vega -

Vega measures the sensitivity of the option price to changes in the volatility of the underlying asset. It is defined as:

$$\nu = \frac{\partial V}{\partial \sigma}$$

While the Black-Scholes model assumes constant volatility, in practice, traders know that volatility is stochastic, and that market-implied volatilities can shift significantly, sometimes more than spot prices do. Vega captures the impact of such changes.

In the Black-Scholes framework, the vega of both calls and puts is the same, and given by:

$$\nu_{call} = \nu_{put} = S \cdot N'(d_1) \cdot \sqrt{T}$$

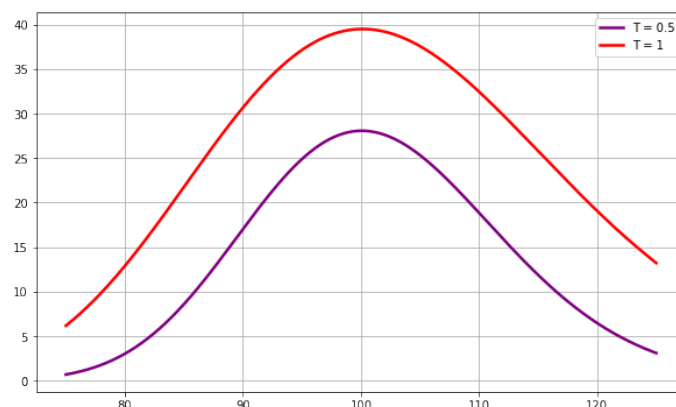


Figure 19: Call vega exposure as a function of the strike

Vega is always *positive* for long vanilla options, because an increase in volatility increases the option's value. This is intuitive: the more uncertain the future, the more valuable the optionality.

Vega is highest when the option is at-the-money, and decreases as the option becomes deep in-the-money or far out-of-the-money — just like gamma. The reason is similar: at-the-money options have the most to gain from increased volatility.

The vega profile is bell-shaped, centered around the at-the-money strike. However, while gamma becomes narrower as maturity decreases (it peaks more), vega does the opposite: its magnitude shrinks. As time to expiry shortens, there's less opportunity for volatility to play out, so vega naturally drops.

Therefore, traders holding short-dated options are generally less exposed to volatility shifts, their greeks are dominated by delta and gamma instead. Conversely, longer-dated options are much more sensitive to changes in implied volatility, and vega management becomes crucial.

In practical trading, gamma and vega exposures are actively monitored and often hedged. This can be done by trading options with different expiries or strikes.

Vega is particularly relevant when implied volatilities move independently of the underlying, such as during earnings announcements, macroeconomic events, or geopolitical shocks.

This "independence" of spot and implied volatility before such scheduled events happens because implied volatility reflects the anticipated magnitude of potential future moves, not the current spot level, so it can rise or fall even if the underlying stays relatively stable.

Theta -

Theta measures the sensitivity of the option price to the passage of time, often referred to as *time decay*. Formally, it is defined as the partial derivative of the

option price with respect to time:

$$\Theta = \frac{\partial V}{\partial t}$$

Since options have an expiration date, time decay is an inevitable factor, especially for options that are out-of-the-money or at-the-money.

In the BS framework, for European options:

$$x = -\frac{S\sigma N'(d_1)}{2\sqrt{T}} \quad y = rKe^{-rT}$$

$$\Theta_{\text{call}} = x - yN(d_2), \quad \Theta_{\text{put}} = x + yN(-d_2)$$

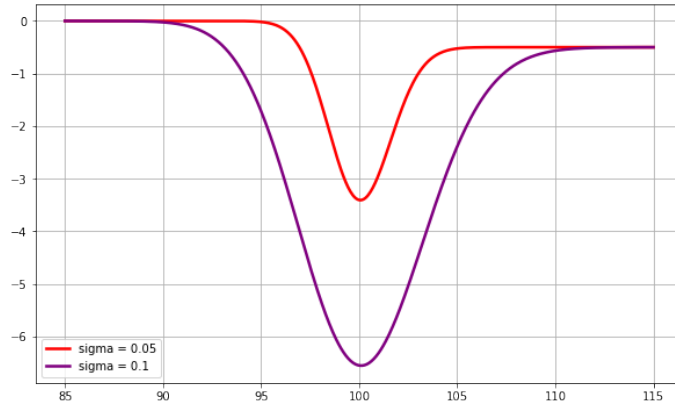


Figure 20: Call theta exposure as a function of the strike

Typically, Θ is negative for long call and put positions, meaning that options lose value as time passes, all else being equal. This loss accelerates as the option approaches maturity, particularly for at-the-money options. Contrarily, option sellers benefit from theta decay, as the premium collected erodes with time.

Understanding theta is crucial for traders managing options positions because it helps quantify how the passage of time impacts profitability.

In summary, theta captures the *cost* of holding options over time, making it a key factor in options trading and risk management strategies.

5.3 Relation between Gamma and Theta -

In the Black-Scholes framework, the value $V(S, t)$ of a European option satisfies the partial differential equation:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0.$$

With Δ , Γ , Θ defined previously, this directly yields:

$$\Theta = -\frac{1}{2}\sigma^2 S^2 \Gamma - rS\Delta + rV.$$

This expression shows that the *gamma* directly contributes to the *theta* through the term:

$$-\frac{1}{2}\sigma^2 S^2 \Gamma,$$

which is always negative for standard long options since $\Gamma > 0$. Therefore, all else being equal, a larger gamma increases the magnitude of the theta. This explains why, near maturity, at-the-money options (which have relatively higher gamma) also exhibit large time decay.

The other terms, $-rS\Delta$ and rV , represent the impact of carrying costs and discounting, respectively, adjusting theta based on interest rates and option price.

In the context of dynamic delta-hedging and gamma management, the theta reflects the P&L drain that offsets the gains from gamma. When gamma is positive, frequent hedge adjustments can generate profits from buying low and selling high in volatile markets, but these gains must first cover the constant bleed from theta.

Conversely, when gamma is negative, the hedging activity tends to lose money in volatile markets. However, this loss can often be offset by positive theta because the option's premium gradually decays in your favor.

Delta determines the immediate P&L resulting from changes in the spot price.

Gamma controls how delta changes over time which influences how frequently and in which direction you need to adjust your hedge.

Theta represents the steady gain or loss due to the passage of time.

Together, these Greeks explain whether a position profits from volatility trading, directional moves, or simply from time decay. In practice, large gamma exposure almost always comes with a large absolute value of theta. This is because the same convexity which is the curvature of the option's price relative to changes in the spot price, where the option price bends rather than moves in a straight line, makes your delta highly sensitive to spot movements and also causes the option's time value to decay more rapidly.

-	Volatility	Δ Sign	Γ Sign	Θ Sign	Δ Magn.	Γ Magn.	Θ Magn.
Buy	High	+	+	-	Large	Large	Large
Sell	High	-	-	+	Large	Large	Large
Buy	Low	+	+	-	Moderate	Moderate	Moderate
Sell	Low	-	-	+	Moderate	Moderate	Moderate

Table 8: Call option greeks sign & magnitude (denoted Magn.) for various directions & market volatility contexts

Quiz (6) - Greeks

Each question has one or many possible answers.

1. At-the-money European options have a strike:
 - ☐ Roughly equal to the underlying spot price
 - ☐ Lower than the spot
 - ☐ Greater than the spot
2. For short-dated at-the-money options, the absolute value of theta:
 - ☐ Remains constant
 - ☐ Becomes very large near expiry
 - ☐ Is negligible
3. Vega is equal for European calls and puts:
 - ☐ Only for at-the-money options
 - ☐ True
 - ☐ False
4. Deep in-the-money call options have delta approximately equal to:
 - ☐ 0
 - ☐ 1
 - ☐ 0.5
5. Delta of a European call option represents:
 - ☐ Sensitivity to volatility
 - ☐ Sensitivity of the option price to changes in the underlying asset price
 - ☐ Sensitivity of the option price to time decay
6. Near expiry, at-the-money options exhibit:
 - ☐ Low gamma and high vega
 - ☐ High gamma and low vega
 - ☐ High gamma and high theta
7. How does gamma behave for at-the-money options as expiry approaches?
 - ☐ Gamma increases
 - ☐ Gamma decreases
 - ☐ Gamma stays constant
8. Does vega directly indicate sensitivity to a volatility skew?
 - ☐ Yes
 - ☐ No
 - ☐ It depends
9. Theta is generally:
 - ☐ Zero at all strikes
 - ☐ Negative for long calls and puts
 - ☐ Positive for long calls and puts
10. Vega is largest in absolute value for options that are:
 - ☐ At-the-money
 - ☐ Deep out-of-the-money
 - ☐ Deep in-the-money

11. Gamma measures:
 - ☐ Sensitivity of option price to interest rate
 - ☐ Sensitivity of delta to the underlying asset price
 - ☐ Time decay
12. Rho of a European call option indicates:
 - ☐ Sensitivity to the underlying asset price
 - ☐ Sensitivity of the option price to volatility
 - ☐ Sensitivity of the option price to interest rate changes
13. What equation links delta, gamma and theta?:
 - ☐ $\Theta = -\frac{1}{2}\sigma^2 S^2 \Gamma - rS\Delta + rV$
 - ☐ $\Theta = -S\Delta + \sigma S^2 \Gamma$
 - ☐ $\Theta = -2r\sigma S^3 \Gamma \Delta V$
14. Long-term options typically have:
 - ☐ Larger vega
 - ☐ Negative gamma
 - ☐ Smaller vega
15. Being short of a European put, your delta is:
 - ☐ Negative
 - ☐ Positive
 - ☐ Approximately 0.5
16. Theta measures:
 - ☐ Sensitivity to volatility
 - ☐ Sensitivity of the option price to interest rate
 - ☐ Sensitivity of the option price to the passage of time
17. Which Greek captures the convexity of the option price with respect to the underlying?
 - ☐ Gamma
 - ☐ Delta
 - ☐ Theta
18. What is the main effect of increasing implied volatility on a call option's price?
 - ☐ Call price increases
 - ☐ Call price stays constant
 - ☐ Call price decreases
19. Which Greek is always positive for a long call and long put?
 - ☐ Theta
 - ☐ Gamma
 - ☐ Rho
20. Vega represents:
 - ☐ Sensitivity of option price to volatility
 - ☐ Sensitivity to the underlying price
 - ☐ Vincent Vega, the character from Pulp Fiction

6 Appendix –

6.1 Itô's lemma

Itô's lemma is a fundamental result in stochastic calculus. It provides the differential of a function of a stochastic process, typically modeled by a Brownian motion or more generally an Itô process. In finance, it is widely used to derive the dynamics of functions of asset prices, such as option pricing formulas.

Let X_t be a 1-dimensional Itô process defined by

$$X_t = X_0 + \int_0^t \mu(s) ds + \int_0^t \sigma(s) dB_s$$

or in differential form:

$$dX_t = \mu(t) dt + \sigma(t) dB_t$$

where $\mu(t)$ and $\sigma(t)$ are *adapted* processes with respect to the Brownian motion B_t

If $f = f(x, t)$ is a function of class $\mathcal{C}^2$ on $\mathbb{R} \times \mathbb{R}_+$, then Itô's lemma states:

$$d(f(X_t, t)) = \frac{\partial f}{\partial t}(X_t, t) dt + \frac{\partial f}{\partial x}(X_t, t) dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(X_t, t) \sigma_t^2 dt.$$

Intuitively, Itô's lemma can be seen as the stochastic analogue of the classical chain rule. In standard calculus, the change of a function $f(x, t)$ is obtained by expanding it in terms of its derivatives with respect to time and space. However, when the variable X_t follows a stochastic process driven by Brownian motion, the situation is different: the paths of Brownian motion are continuous but nowhere differentiable.

When performing a Taylor expansion of $f(X_t, t)$ over a small time step, one finds that the random increment dB_t cannot be treated as an infinitesimal of higher order. In fact, its square $(dB_t)^2$ contributes on the order of dt . This phenomenon forces the appearance of an additional correction term in the expansion.

As a result, in addition to the usual first-order derivatives with respect to t and x , Itô's lemma includes a second-order term involving the second derivative of f with respect to x . This term captures the accumulated variance of the stochastic fluctuations, which is the essential difference between stochastic and deterministic calculus.

6.2 S_T analytic expression

Starting from the BS SDE:

$$dS_t = \mu S_t dt + \sigma S_t dW_t \quad (1)$$

where $\mu \in \mathbb{R}$, $\sigma > 0$, and $(W_t)_{t \geq 0}$ is a standard Brownian motion.

Applying Itô's lemma to $f(S_t, t) = \ln(S_t)$ with $X_t = \ln S_t$

$$dX_t = \frac{1}{S_t} dS_t - \frac{1}{2} \frac{1}{S_t^2} (dS_t)^2. \quad (2)$$

Squaring (1) using the $(a + b)^2$ formula, and knowing that: $dW_t^2 = dt$, $(d_t)^2 = 0$ and $d_t dW_t = 0$

$$(dS_t)^2 = (\sigma S_t dW_t)^2 = \sigma^2 S_t^2 dt \quad (3)$$

Using (1) to replace the first term, and substituting (2)

$$dX_t = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t$$

$$\int_0^T dX_t = \int_0^T \left(\mu - \frac{1}{2}\sigma^2\right) dt + \int_0^T \sigma dW_t$$

$$X_T - X_0 = \left(\mu - \frac{1}{2}\sigma^2\right) T + \sigma W_T$$

$$X_T = \ln S_0 + \left(\mu - \frac{1}{2}\sigma^2\right) T + \sigma W_T$$

Exponentiating both sides:

$$S_T = e^{X_T} = S_0 e^{(\mu - \frac{1}{2}\sigma^2)T + \sigma W_T}$$

6.3 Black-Scholes and the Heat Equation

As seen previously, the price of a European call or put option $V(S, t)$, as a function of the underlying asset price S and time t , satisfies the PDE:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

Consider the change of variables:

$$x = \ln \frac{S}{K}, \quad \tau = T - t, \quad V(S, t) = Ke^{-r\tau} u(x, \tau)$$

Compute derivatives

$$\frac{\partial V}{\partial t} = -Ke^{-r\tau} \frac{\partial u}{\partial \tau} - rKe^{-r\tau} u$$

$$\frac{\partial V}{\partial S} = \frac{Ke^{-r\tau}}{S} \frac{\partial u}{\partial x}$$

$$\frac{\partial^2 V}{\partial S^2} = \frac{Ke^{-r\tau}}{S^2} \left(\frac{\partial^2 u}{\partial x^2} - \frac{\partial u}{\partial x} \right)$$

Substitute into the BS PDE

$$\begin{aligned} & \left(-Ke^{-r\tau} \frac{\partial u}{\partial \tau} - rKe^{-r\tau} u \right) + \frac{1}{2}\sigma^2 S^2 \cdot \frac{Ke^{-r\tau}}{S^2} \left(\frac{\partial^2 u}{\partial x^2} - \frac{\partial u}{\partial x} \right) \\ & + rS \cdot \frac{Ke^{-r\tau}}{S} \frac{\partial u}{\partial x} - rKe^{-r\tau} u = 0 \end{aligned}$$

Factor $Ke^{-r\tau}$ and simplify:

$$-\frac{\partial u}{\partial \tau} + \frac{1}{2}\sigma^2 \frac{\partial^2 u}{\partial x^2} + \left(r - \frac{1}{2}\sigma^2 \right) \frac{\partial u}{\partial x} - ru = 0$$

Remove the first-order derivative Finally, to reduce it to the standard heat equation, apply a further transformation:

$$u(x, \tau) = e^{\alpha x + \beta \tau} w(x, \tau)$$

with appropriate constants α, β chosen to eliminate the $\frac{\partial w}{\partial x}$ and constant terms. After some algebra, one obtains the classical heat equation:

$$\frac{\partial w}{\partial \tau} = \frac{1}{2}\sigma^2 \frac{\partial^2 w}{\partial x^2}$$

This link between the BS PDE and the heat equation provides a deep insight: option pricing can be seen as a diffusion process in "log-price space" analogous to the diffusion of heat in a rod. This analogy allows the use of powerful analytical and numerical techniques from physics to solve financial problems efficiently.

6.4 Other useful option contracts

6.4.1 American Option

An *American option* is a type of option contract that gives its holder the right, but not the obligation, to buy or sell the underlying asset at a specified strike price at any time up to and including the expiration date.

The payoff of an American call or put option at exercise time $\tau \leq T$ can be written as:

$$\max(S_\tau - K, 0) \quad \text{or} \quad \max(K - S_\tau, 0)$$

where S_τ is the price of the underlying at the exercise time τ and K is the strike price.

The holder can choose τ as he wants, with $0 \leq \tau \leq T$

This flexibility distinguishes American options from European options, which can only be exercised at maturity. The early-exercise feature introduces additional complexity in pricing, as there is no analytical formula to price them in a generic case. Numerical methods such as the binomial tree, finite difference methods, or Monte Carlo simulations are commonly used. These methods evaluate the option's value at some potential exercise times.

American options are common in equity markets, where investors may wish to exercise the option prior to an ex-dividend date or to capture favorable movements in the underlying asset.

6.4.2 Digital Option

A *digital option* (also called a binary option) is a type of option contract that pays a fixed amount if the underlying asset satisfies a certain condition at expiration, and nothing otherwise.

The payoff of a digital call or put option at maturity T can be written as:

$$\mathbf{1}_{\{S_T > K\}} \quad \text{or} \quad \mathbf{1}_{\{S_T < K\}}$$

where S_T is the price of the underlying at maturity and K is the strike price.

With $\mathbf{1}_{\{S_T > K\}}$ denoting the indicator function, defined as

$$\mathbf{1}_x = \begin{cases} 1 & \text{if } x \\ 0 & \text{otherwise} \end{cases}$$

In the Black-Scholes model, the digital call and put can be priced analytically.

$$C_{\text{digital}} = e^{-rT} N(d_2)$$

$$P_{\text{digital}} = e^{-rT} N(-d_2)$$

6.5 Monte-Carlo variance reduction methods

Monte-Carlo simulation is widely used in a variety of problems. However, it may be inefficient due to high variance, requiring a large number of simulations to achieve accurate estimates.

To see why high variance is problematic, consider a Monte-Carlo estimator for an expectation $\mathbb{E}(X)$:

$$\hat{\mu}_N = \frac{1}{N} \sum_{i=1}^N X_i \quad (1)$$

where $(X_i)_{i \in 1 \dots N}$ are i.i.d. sampling the random variable X . The variance of this estimator is

$$\text{Var}(\hat{\mu}_N) = \frac{\text{Var}(X)}{N}.$$

This shows that the standard error (square root of the variance) decreases only as $1/\sqrt{N}$. If $\text{Var}(X)$ is large, achieving a desired accuracy requires a very large number of simulations, which can be computationally expensive. Variance reduction techniques aim to lower $\text{Var}(X)$, improving the efficiency of the estimator without increasing N .

Variance reduction techniques aim to improve the efficiency of Monte-Carlo by reducing the estimator's variance without increasing the number of simulations.

Common variance reduction methods include:

- **Antithetic Variates:** For each simulated variable, also simulate a mirrored version (e.g. using $-Z$ if Z is a standard normal). Averaging the results reduces variance.
- **Control Variates:** Use a related variable with a known expected value to reduce the variance of the estimator. For example, if pricing an exotic option, use a plain European option as a control.
- **Importance Sampling:** Change the probability distribution to sample more frequently in regions that contribute most to the payoff, then weight the samples appropriately.

Let's prove the first method "antithetic variates" successfully decreases the variance.

Antithetic Variates: Let's apply this technique in the situation (1) described above.

For each sample X_i we also generate a dependent sample $X'_i = f(X_i)$ such that $\mathbb{E}[X'_i] = \mathbb{E}[X_i] = \mu$, often using $X'_i \stackrel{\mathcal{L}}{=} -X_i$ for symmetric distributions like the normal law. We define the antithetic estimator:

$$\hat{\mu}_N^{\text{anti}} = \frac{1}{2N} \sum_{i=1}^N (X_i + X'_i)$$

The variance of this estimator is

$$\text{Var}[\hat{\mu}_N^{\text{anti}}] = \frac{1}{4N} \text{Var}[X_i + X'_i].$$

Expanding $\text{Var}[X_i + X'_i]$:

$$\text{Var}[X_i + X'_i] = \text{Var}[X_i] + \text{Var}[X'_i] + 2 \text{Cov}[X_i, X'_i].$$

If X'_i is negatively correlated with X_i i.e. $\text{Cov}[X_i, X'_i] < 0$, then

$$\text{Var}[X_i + X'_i] < \text{Var}[X_i] + \text{Var}[X'_i] = 2 \text{Var}[X_i].$$

Thus

$$\text{Var}[\hat{\mu}_N^{\text{anti}}] = \frac{1}{4N} \text{Var}[X_i + X'_i] < \frac{1}{4N} \cdot 2 \text{Var}[X_i] = \frac{\text{Var}[X_i]}{2N} < \text{Var}[\hat{\mu}_N],$$

showing that the antithetic estimator has lower variance than the standard Monte-Carlo estimator.

6.6 Example of a more sophisticated diffusion model

Unlike the BS model, which assumes constant volatility and allows for closed-form option pricing, many more realistic models feature *stochastic volatility*. These models capture the empirical fact that implied volatilities vary across strikes and maturities.

A canonical example is the **Heston model**, theorized by Steven Heston, where the underlying asset S_t and its variance v_t evolve as:

$$\begin{cases} dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t^S, \\ dv_t = \kappa(\theta - v_t) dt + \sigma \sqrt{v_t} dW_t^v, \end{cases}$$

with $\text{Cov}(dW_t^S, dW_t^v) = \rho dt$

- κ is the mean-reversion speed of variance
- θ is the long-term variance level
- σ is the volatility of the variance process ("volatility of volatility")
- ρ is the correlation between the two brownian motions W^S and W^v

No closed-form formula for option pricing. Monte-Carlo simulation or numerical PDE methods are required. Indeed, in the BS model, S_T is lognormally distributed, so the European call price can be computed exactly using the cumulative normal distribution.

In the Heston model, the variance v_t is itself stochastic. Because v_t is random and evolves over time, the distribution of S_T conditional on S_0 depends on the entire path of v_t . Unlike the lognormal case, there is no simple formula for the density of S_T :

$$S_T = S_0 e^{\int_0^T \left(r - \frac{1}{2} v_s\right) ds + \int_0^T \sqrt{v_s} dW_s^S}.$$

Thus, the expectation

$$C(S_0, 0) = e^{-rT} \mathbb{E}[(S_T - K)^+]$$

involves a nontrivial integral over all possible paths of v_t . While the Heston model allows a semi-analytical solution for European options using characteristic functions and Fourier inversion (via the *complex-valued* conditional moment-generating function of $\ln S_T$), there is no closed-form formula in terms of elementary functions.

Glossary

Alpha A measure of an investment's performance relative to a benchmark, representing excess return due to skill, strategy or luck.

Ask The price at which a seller is willing to sell a security or asset.

Bankruptcy The legal state of a company or individual unable to meet its financial obligations, leading to debt restructuring or liquidation.

Beta A measure of an asset's sensitivity to movements in the overall market, representing systematic risk exposure.

Bid The price at which a buyer is willing to purchase a security or asset.

Broker A person or firm that acts as an intermediary between buyers and sellers of financial securities, facilitating trades, providing market access, liquidity, and sometimes investment advice, typically earning a commission for executing trades.

Clearing The process of settling trades and ensuring the transfer of funds and securities between buyers and sellers (cf settlement).

Compilation The process of translating source code written in a programming language into executable machine code or bytecode.

Correlation A statistical measure of the strength and direction of the linear relationship between two random variables, ranging from -1 to 1.

Derivative A financial instrument whose value is derived from the performance of an underlying asset, index, or rate.

Diversification The practice of spreading investments across different assets or sectors to reduce risk.

Equity The ownership interest in a company, usually represented by shares or stock.

ESG Environmental, Social, and Governance criteria used to evaluate a company's ethical and sustainable practices.

Exposure The potential risk of loss or gain in an investment or portfolio due to changes in market prices, rates, volatility, time passing or other factors.

Fair Price The theoretically correct value of a financial instrument.

Forward A customized contract obligating the purchase or sale of an asset at a predetermined price on a future date, traded over-the-counter.

Future A standardized contract to buy or sell an asset at a predetermined price and date, traded on an exchange (cf standardized).

GDP Gross Domestic Product: the total monetary value of all goods and services produced within a country over a specific period.

Idiosyncratic Risk The portion of risk specific to a single asset or small group of assets, which can be reduced through diversification.

Index Statistical measure of the performance of a group of assets, often representing a market or sector.

Inflation The rate at which the general level of prices for goods and services rises, eroding purchasing power.

Leveraged Referring to an investment or position that is financed with borrowed funds, amplifying both potential gains and potential losses.

Liquidation The process of closing a business or investment position by selling assets, typically to pay off debts or to convert holdings into cash.

Liquidity The ease with which an asset can be bought or sold in the market without affecting its price.

Margin Call A broker's demand for an investor to deposit additional funds or securities to cover potential losses on a leveraged position.

Market Making The practice of continuously quoting buy and sell prices for a financial instrument to provide liquidity in the market (c.f. leveraged).

Notional The total nominal value of a financial instrument, often used in derivatives to calculate payments without representing actual cash exchanged.

OTC (Over-the-counter) a decentralized market where financial instruments such as stocks, bonds, derivatives, or currencies are traded directly between parties, rather than through a centralized exchange.

P&L Profit and Loss: the measure of financial gains or losses over a specific period.

Premium A price defined for a financial contract, asset, or service, often representing compensation for risk, rights, or benefits beyond the nominal value.

Primary Market The market where new securities are issued and sold for the first time, typically via IPOs or bond issuances.

Proprietary (PROP) Trading dealing financial instruments using a firm's own capital, rather than client funds, to generate profits.

Quotation The latest market price at which a security is being offered for sale or purchase.

- Secondary Market** The market where previously issued securities are traded among investors, such as stock exchanges.
- Settlement** The process of completing a trade by transferring the asset and corresponding payment(s) between buyer and seller.
- Standardized** Designed or structured according to uniform specifications or rules, allowing for consistency, comparability, and ease of trading in financial markets.
- Stochastic** Randomly determined; involving a sequence of random variables or processes.
- Sustainability** The practice of managing economic, social, and environmental impacts to ensure long-term viability and responsibility.
- Swap** A derivative contract in which two parties exchange cash flows or liabilities from different financial instruments, often to hedge risk or manage exposure.
- SX5E** The ticker symbol for the EURO STOXX 50 index, representing the 50 largest and most liquid stocks in the Eurozone.
- Systematic Risk** The portion of market risk that affects all assets and cannot be eliminated through diversification, e.g. interest rate or economic risk.
- Underlying** The financial asset (e.g. stock, bond, commodity) on which a derivative contract is based.
- Volatility** A measure of the degree of variation of a financial instrument's price over time, often used as a proxy for risk.
- Yield** The income return on an investment, usually expressed as a percentage of the investment's cost or current market value.

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